SAKSHI PARATE - 7073022

Single Cell Project 2

Week 1

1 Preprocessing and quality control

1.1 Set up the environment

The required datasets were downloaded from the provided website and prepared for use in the analysis pipeline. The reference genome was set to **hg38**, with parallel processing enabled where supported.

1.2 Read the data into an appropriate data structure and apply filtering

The downloaded fragment files were imported into Arrow files, and lenient quality filtering was applied using thresholds of **filterFrags = 550** and **filterTSS = 5**.

1.3 Identify doublets

Doublet identification was performed, and detected doublets were removed from the dataset, reducing the cell count from approximately **9500** cells to **6073**. This indicates that roughly 3400 cells were classified as doublets or doublet like and excluded from further analysis.

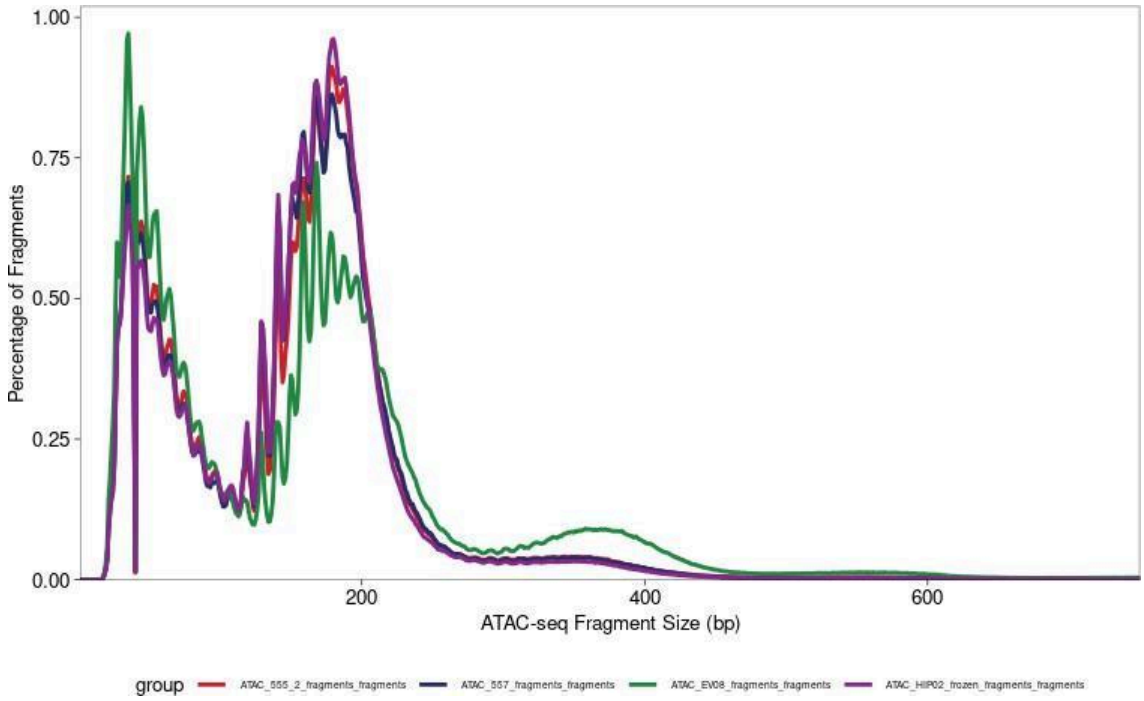
1.4 Collect all samples into a joint data structure

All samples were merged into a joint ArchR project containing **9,460 cells** before doublet removal.
The median **TSS enrichment** in the dataset was **9.56**.
The median **number of fragments** was **5,577**.
The tile matrix had dimensions of **6,062,095 × 9,460** (tiles × cells).

1.5 Quality control

How many cells per sample?
ATAC_555_2 → **1,227** cells.
ATAC_557 → **851** cells.
ATAC_EV08 → **970** cells.
ATAC_HIP02_frozen → **6,412** cells.

Plot the fragment length distribution of all samples in a single plot.



Explain the biological significance of the periodicity observed in the fragment length distribution. What do the peaks at < 100 bp,~ 200 bp, and~ 400 bp represent physically in the chromatin context?

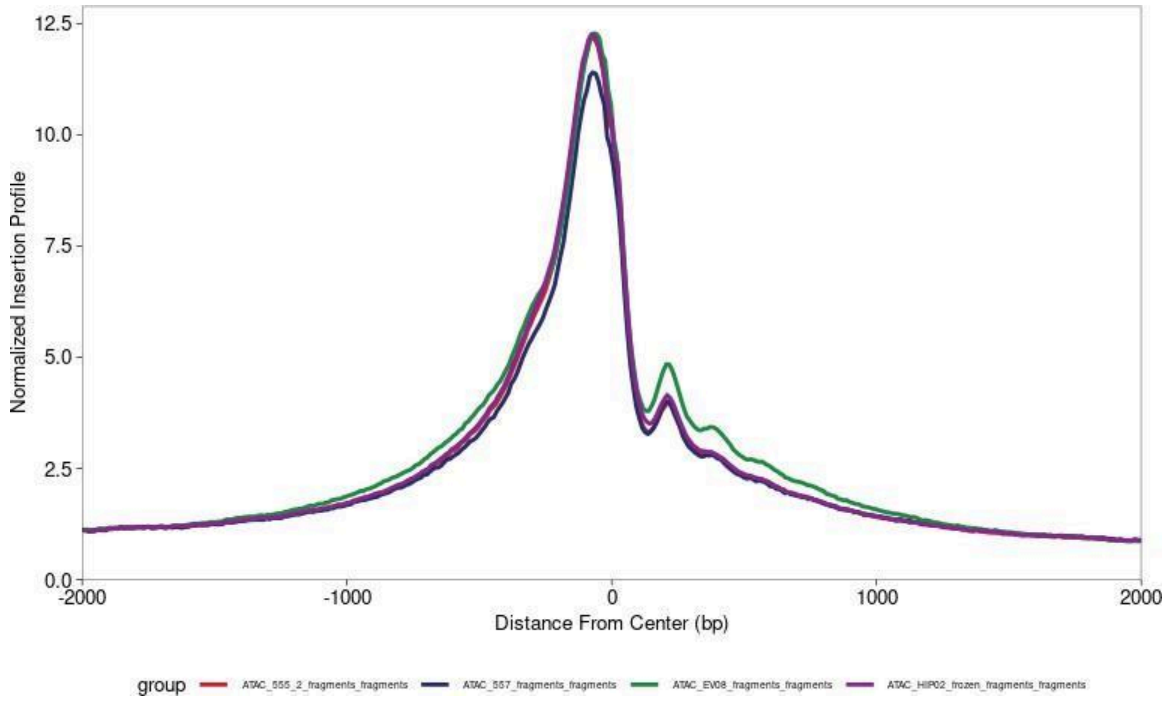
The peak at **<100 bp** corresponds to **nucleosome free fragments** (open chromatin at promoters/enhancers).

The **~200 bp** peak corresponds to **mononucleosomal fragments** (DNA protected by a single nucleosome).

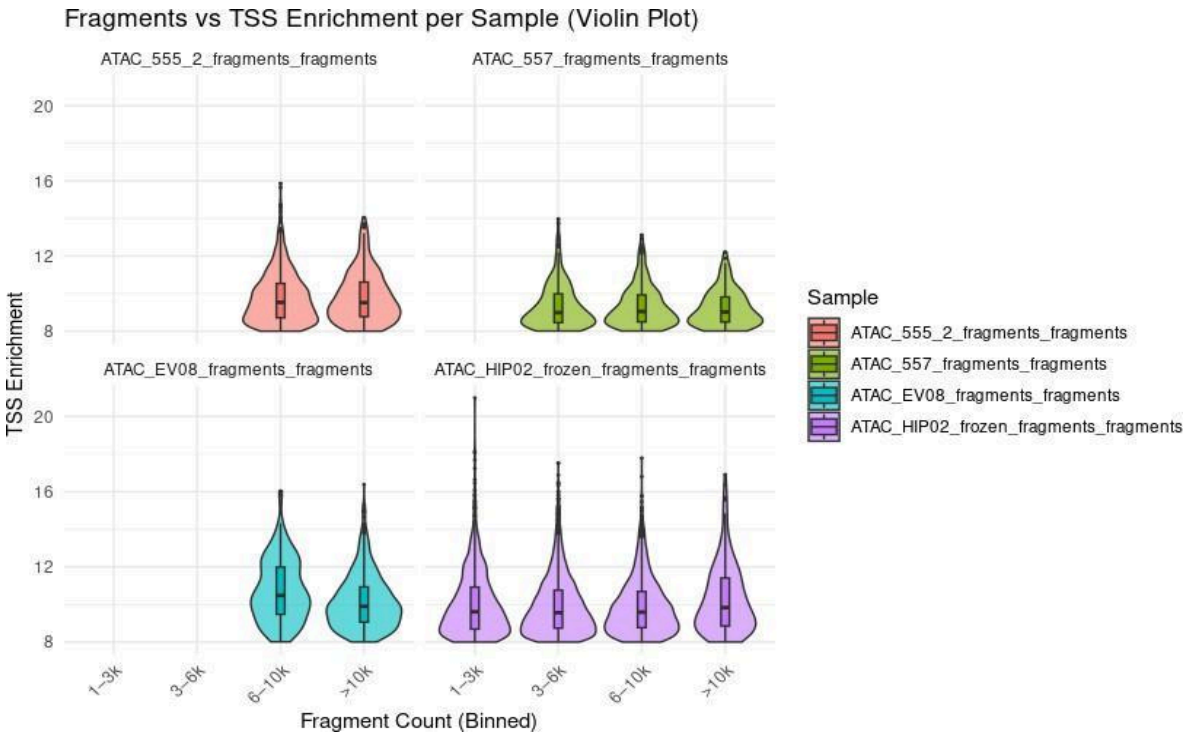
The **~400 bp** peak corresponds to **dinucleosomal fragments** (DNA protected by two adjacent nucleosomes).

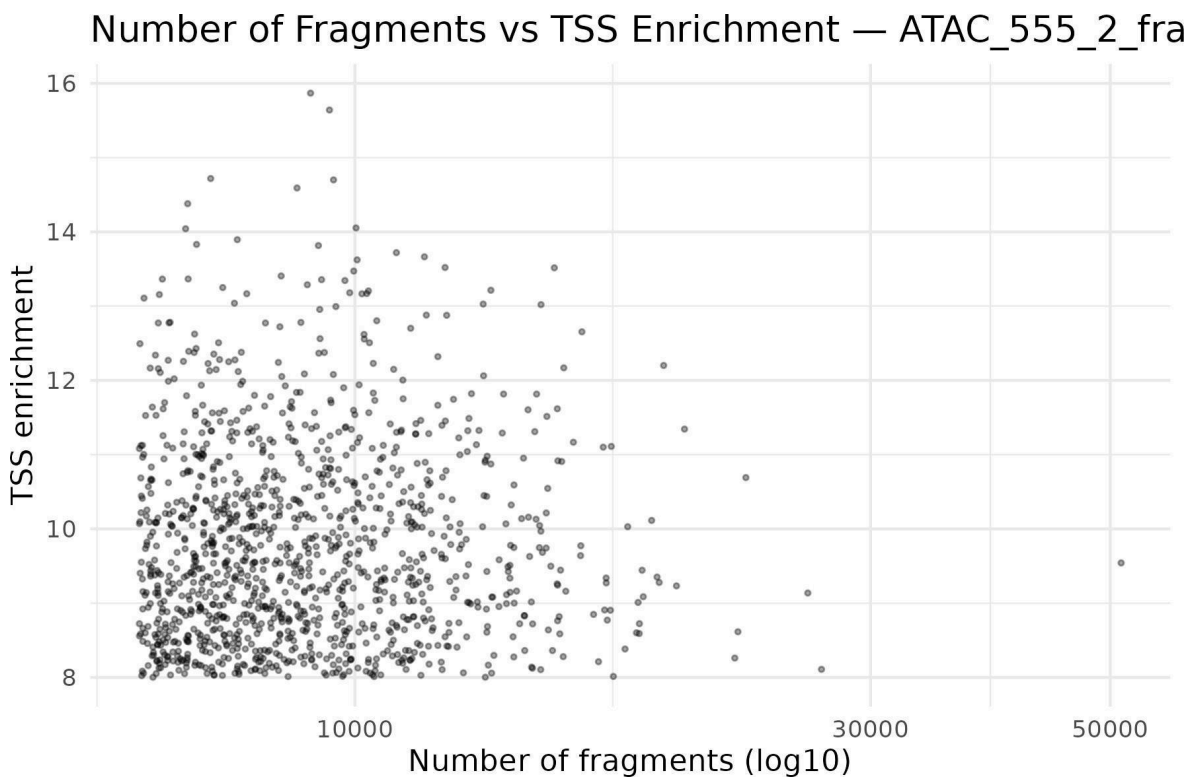
These periodic peaks indicate preserved nucleosome organization and overall good library quality.

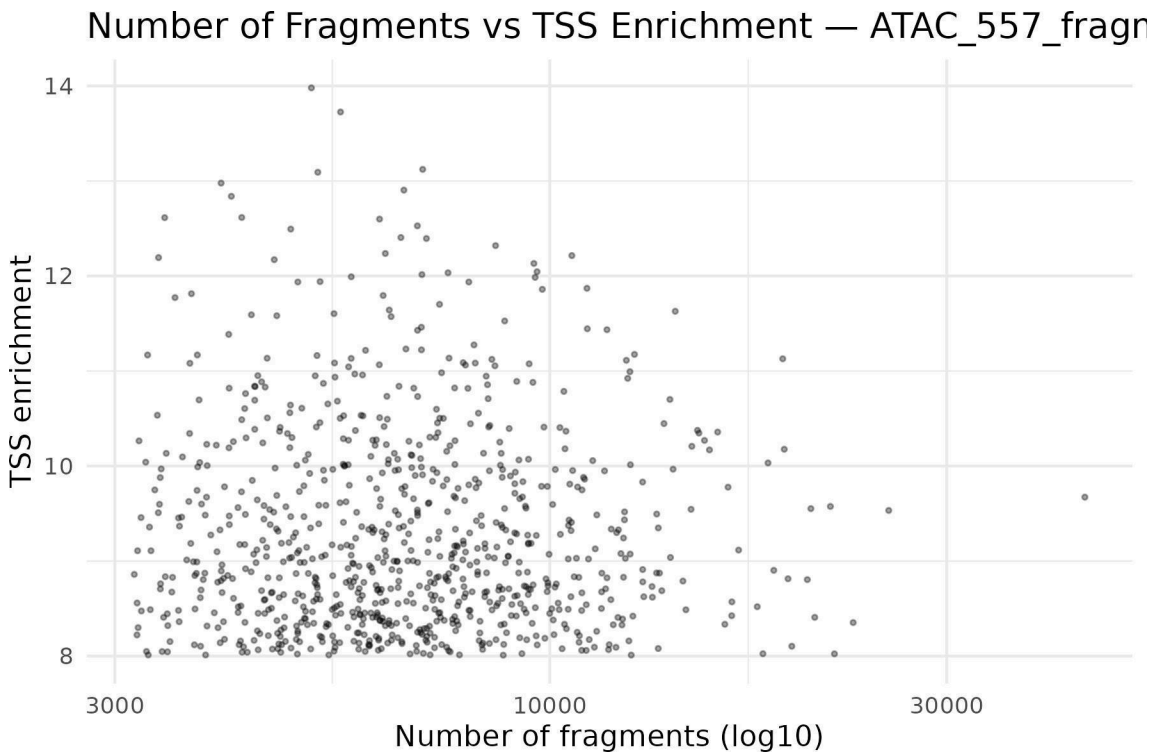
Plot the distribution of TSS enrichment scores in each sample.

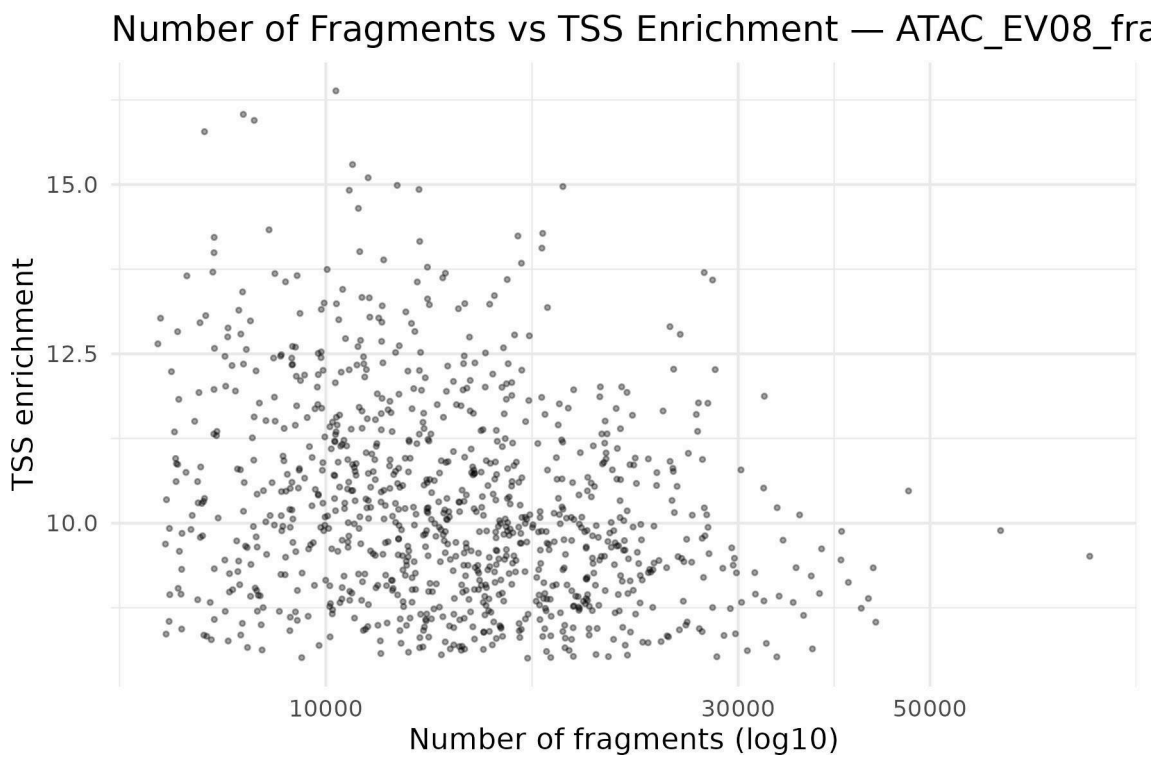


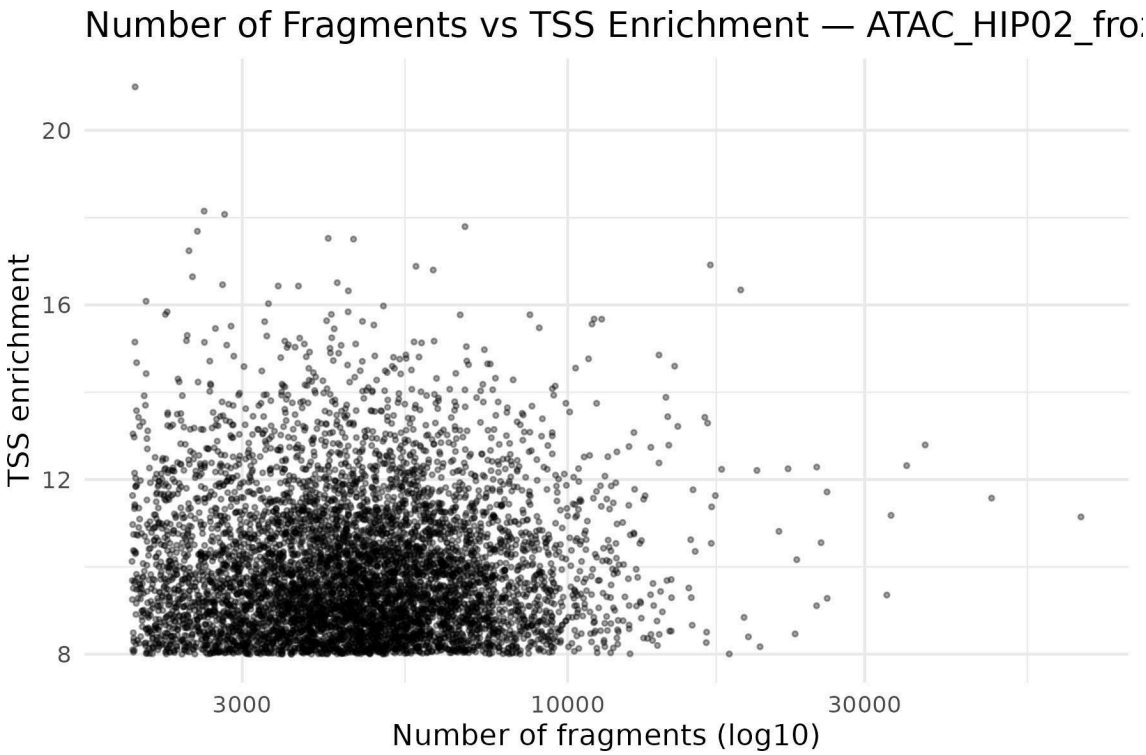
For each sample, plot the number of fragments vs TSS enrichments.




Number of Fragments vs TSS Enrichment — ATAC_555_2_fra
TSS enrichment
16
14
12
10
8
10000
30000
50000
Number of fragments (log10)


Number of Fragments vs TSS Enrichment — ATAC_557_fragn
TSS enrichment
14
12
10
8
3000
10000
30000
Number of fragments (log10)


Number of Fragments vs TSS Enrichment — ATAC_EV08_fra
TSS enrichment
15.0
12.5
10.0
10000
30000
50000
Number of fragments (log10)


Number of Fragments vs TSS Enrichment — ATAC_HIP02_fro
TSS enrichment
20
16
12
8
3000
10000
30000
Number of fragments (log10)

Inspect the plots and report your findings. Do you observe outlier samples/samples with low quality?

No samples were identified as clear outliers or of low quality. Fragment length periodicity, TSS enrichment profiles, and fragment count vs TSS relationships are consistent across samples; ATAC_HIP02_frozen shows more points due to higher cell count but follows the same quality pattern.

1.5 Filter the dataset

Stricter filtering was applied based on fragment counts, TSS enrichment, and removal of doublets.

Number of cells in the filtered project: **5,089** cells.

Median TSS enrichment (after filtering): **9.525**.

Median number of fragments (after filtering): **5,108**.

Tile matrix dimensions after filtering: **6,062,095 × 5,089** (tiles × cells).

Week 2

2 Peaks

2.1 Peak calling

1. Explain what was done and why this is necessary
Peak calling was performed using the strictly filtered ArchR project, where low quality cells and predicted doublets had already been removed based on fragment counts, TSS enrichment, and doublet scores. Cells were then grouped by sample to generate pseudo bulk replicates, which aggregate accessibility signals across similar cells. This preparation is necessary because individual single cell ATACseq profiles are extremely sparse and do not provide sufficient read depth for reliable peak detection by tools such as MACS2.

2. Why call peaks on pseudo-bulk replicates rather than single cells or the total merged dataset? How does this approach help identify rare cell-type-specific elements?

Single cells are too sparse, leading to noisy and unreliable peak calls. Calling peaks on the total merged dataset favors abundant cell populations and can mask regulatory elements specific to rare cell types. Pseudo bulking by samples or clusters preserves biological heterogeneity while increasing signal to noise ratio. This strategy improves the detection of rare cell type specific regulatory elements, as peaks present in small cell populations are strengthened within their corresponding pseudo bulk groups instead of being diluted.

3. Compute a joint peak set and add the peak matrix
A joint reproducible peak set was computed from the pseudo-bulked groups. The resulting peak set was added to the ArchR project, and a peak by cell accessibility matrix was generated for downstream analyses.

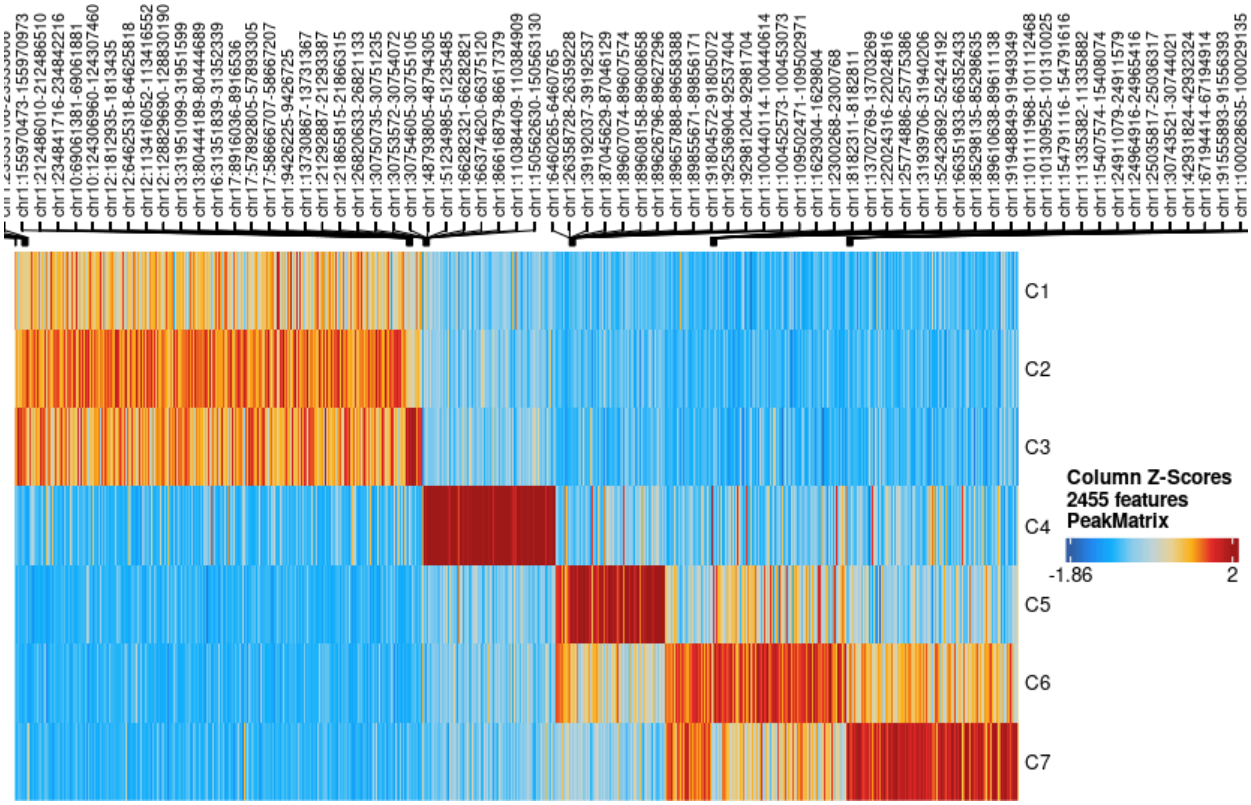
2.1 Cluster marker peaks

Which parameters are used for which variable and what do they describe?

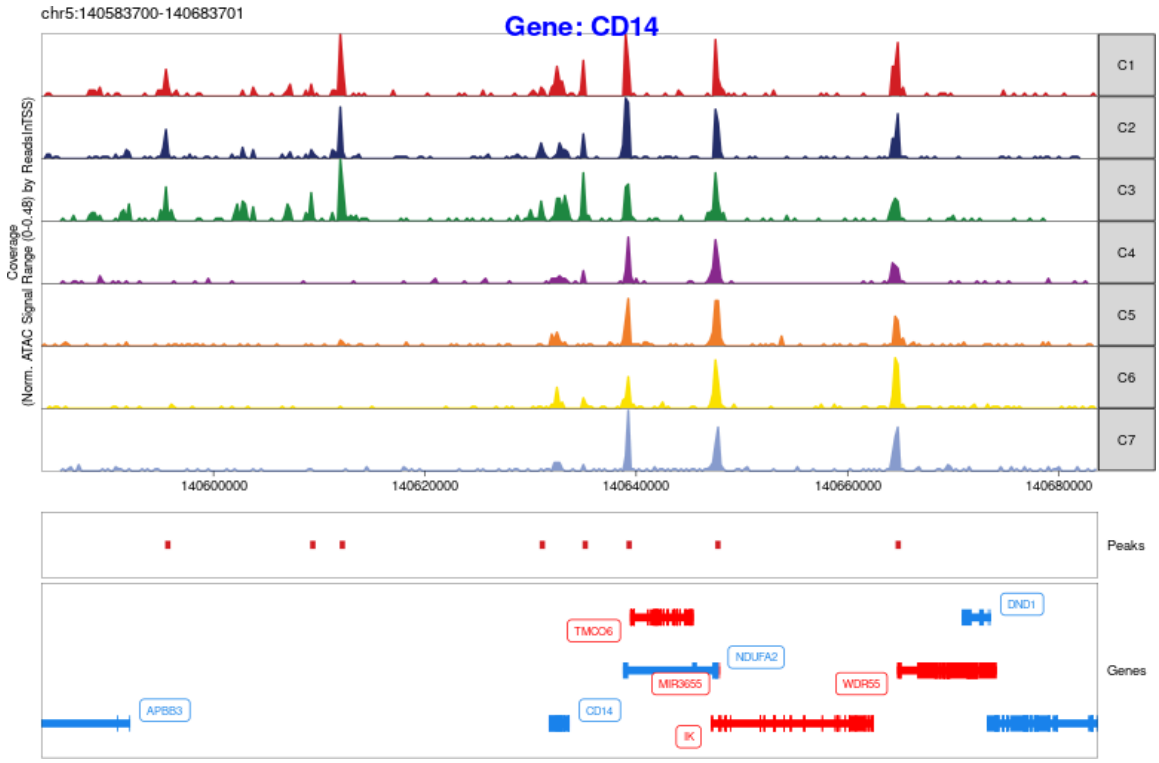
- useMatrix = “PeakMatrix” - Uses peak-level chromatin accessibility values for differential testing.
- groupBy = “Clusters” - Defines the biological grouping, testing accessibility differences between clusters.

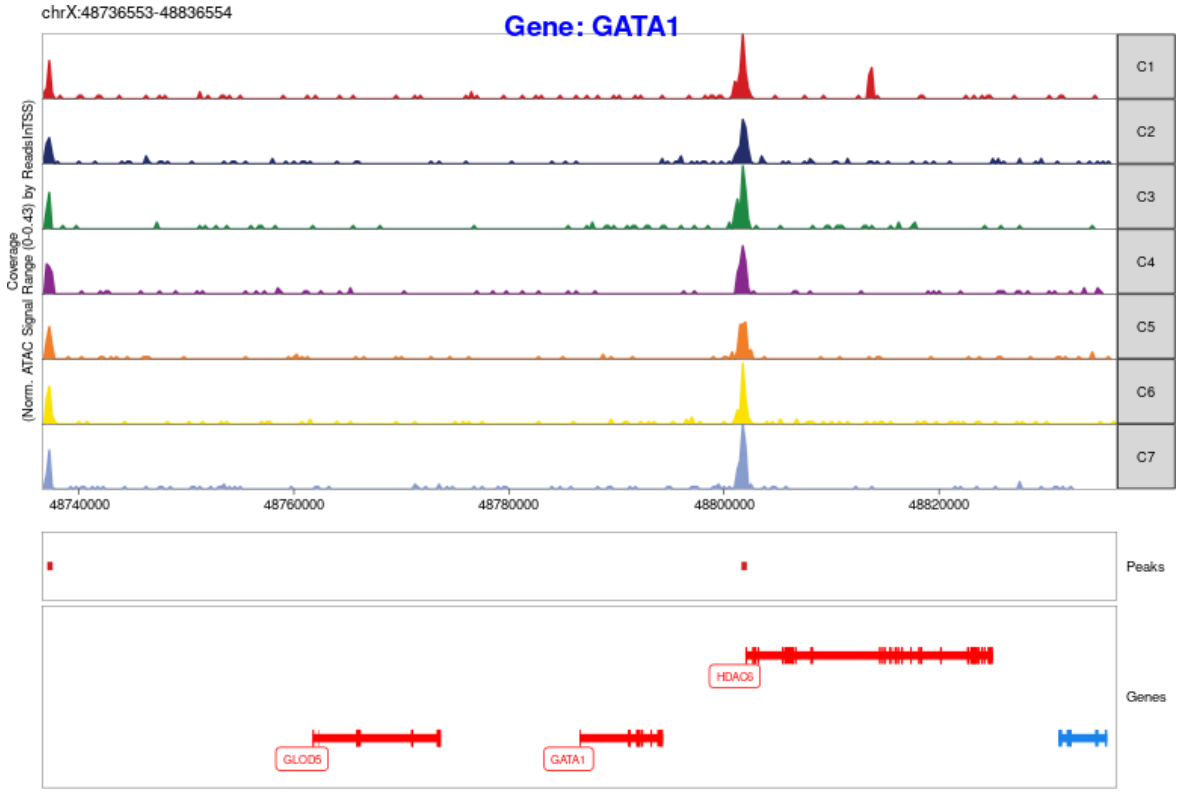
- testMethod = “wilcoxon” - A non parametric statistical test suitable for sparse single cell ATAC-seq data.
- bias = c(“TSSEnrichment”, “log10(nFrags)”) - Corrects for technical confounders:
  TSSEnrichment accounts for chromatin signal quality.
  nFrags accounts for sequencing depth per cell.
- Marker peak cutoffs (applied post-testing):
  FDR ≤ 0.01 ensures statisticalsignificance.
  Log2FC ≥ 1 ensures biologically meaningful differences in accessibility.

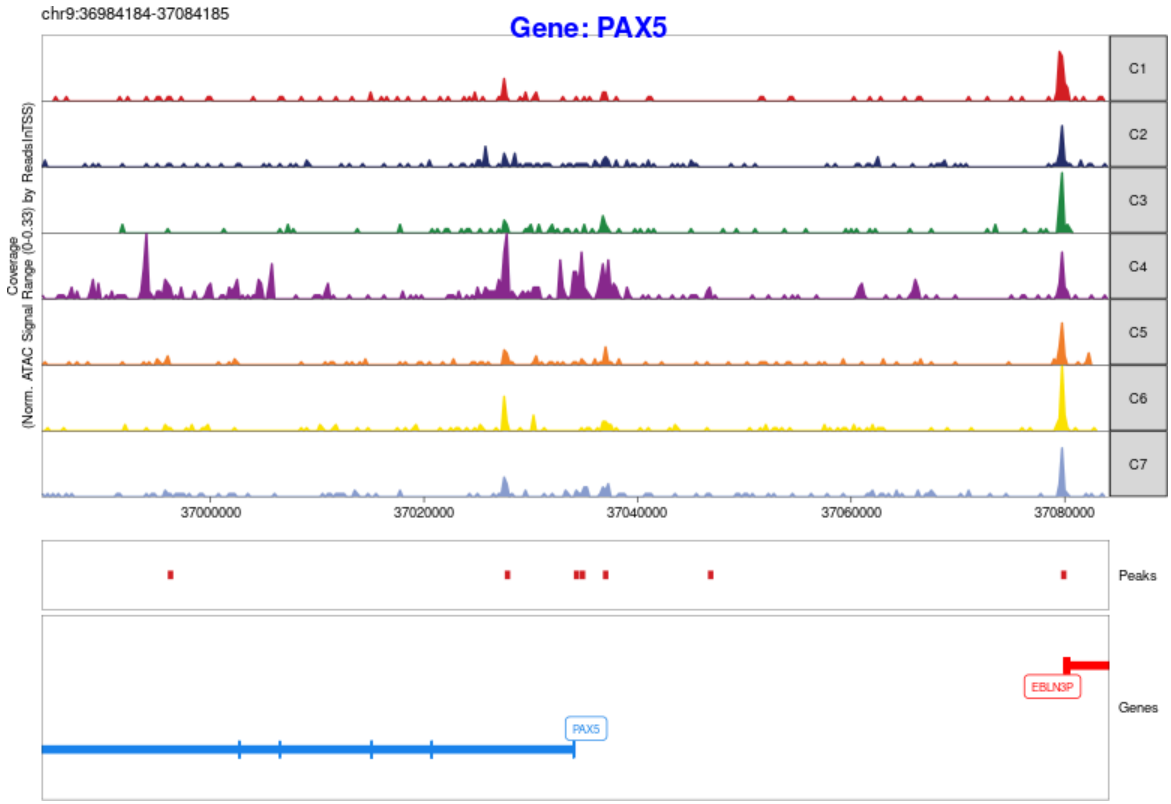
Create a Heatmap showing accessibility in the marker peaks across the clusters.

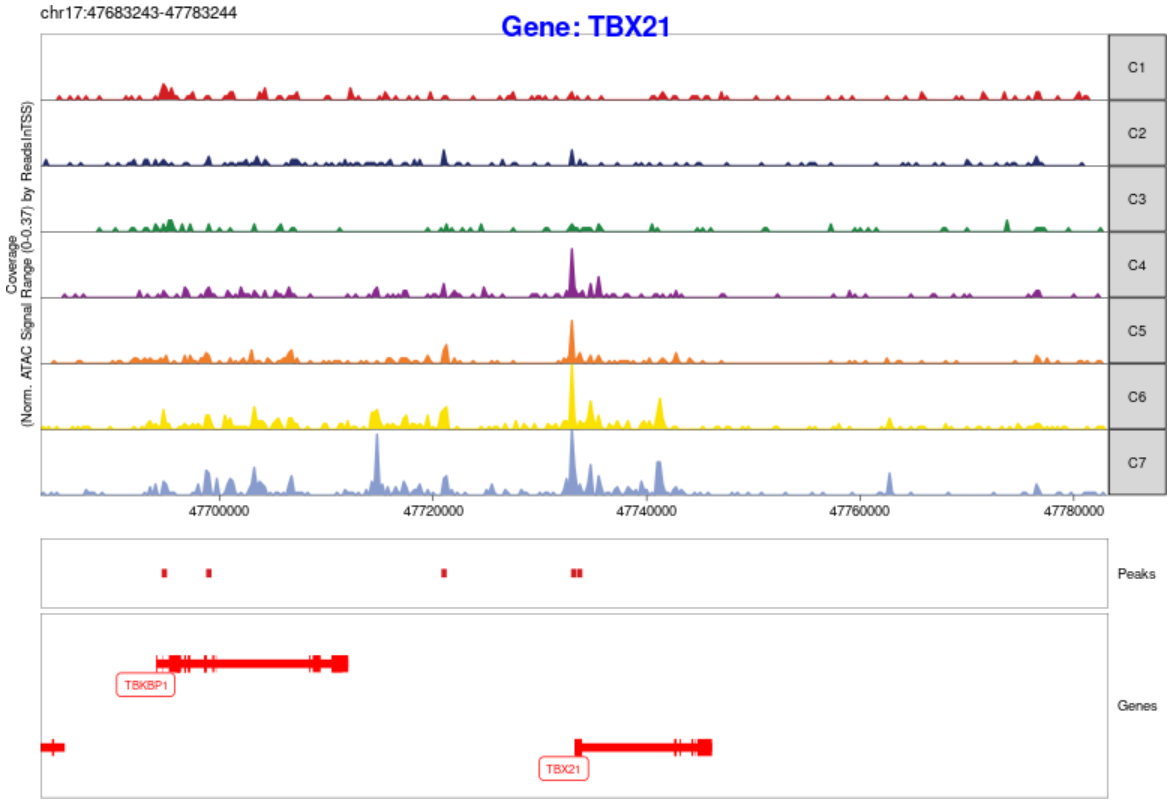


Create plots around the following genes: CD8A, CD14, GATA1, PAX5 , TBX21.




chrX:48736553-48836554
Gene: GATA1
Coverage
(Norm. ATAC Signal Range (0-0.43) by ReadsInTSS)
C1
C2
C3
C4
C5
C6
C7
48740000
48760000
48780000
48800000
48820000
Peaks
Genes
HDAC6
GLOD5
GATA1


chr9:36984184-37084185
Gene: PAX5
Coverage
(Norm. ATAC Signal Range (0-0.33) by ReadsInTSS)
C1
C2
C3
C4
C5
C6
C7
37000000
37020000
37040000
37060000
37080000
Peaks
Genes
EBLN3P
PAX5



3 Dimensionality Reduction

3.1 Iterative LSI

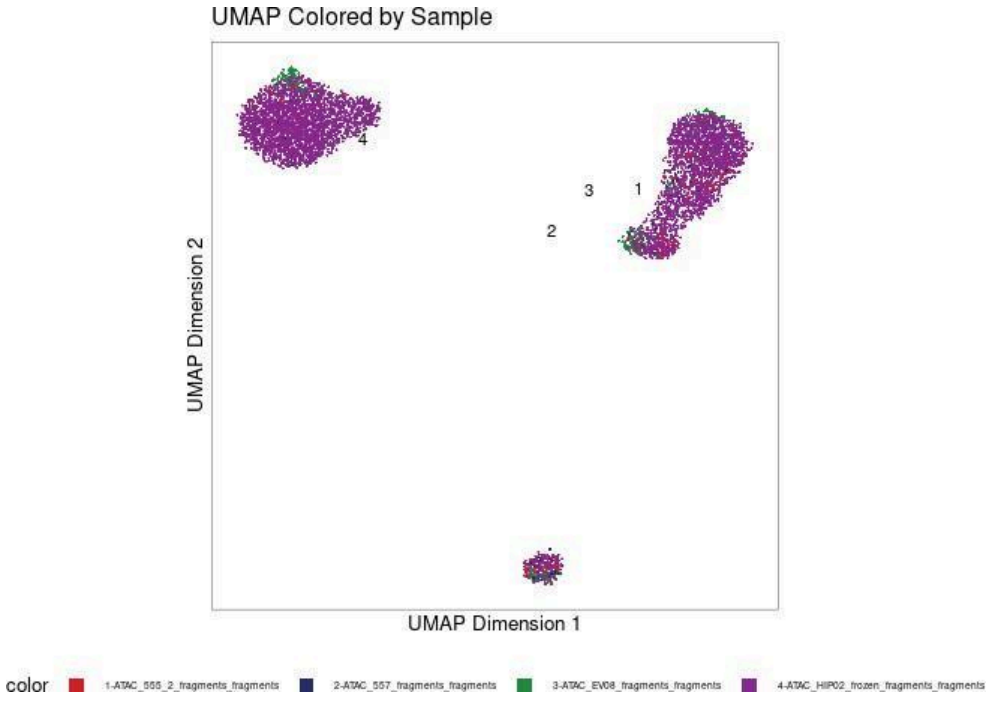
1. Why do we use LSI and not only PCA for the dimensionality reduction of scATAC data?

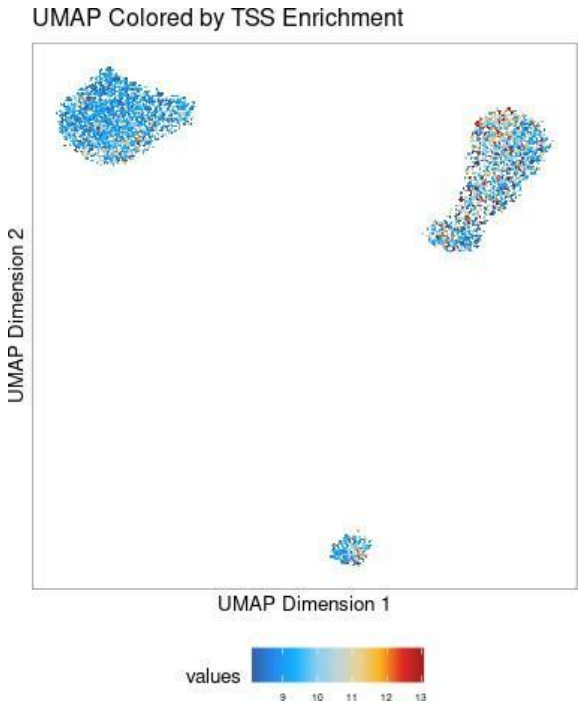
scATACseq data is highly sparse and largely binary like, with most peaks being inaccessible in most cells. PCA assumes continuous, normally distributed data and therefore does not model scATACseq data well. LSI was originally developed for sparse count data (e.g. text mining) and is well suited for peak accessibility matrices. LSI reduces the influence of sequencing depth and technical noise while capturing meaningful chromatin accessibility patterns

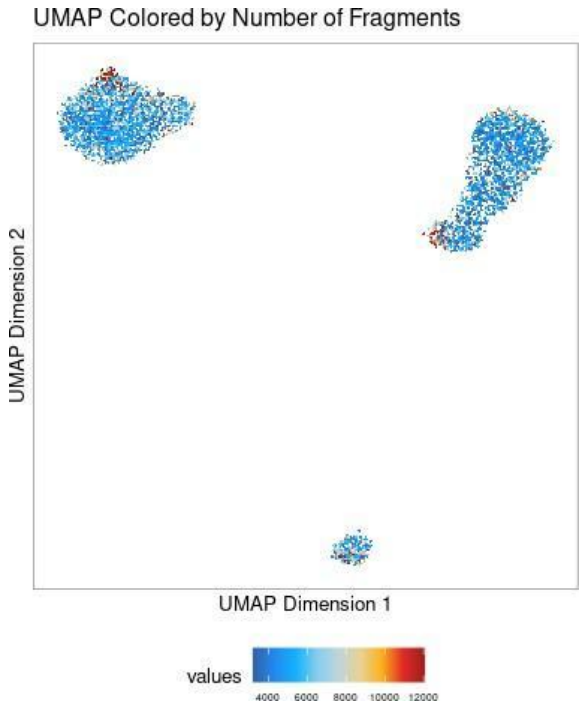
2. Why is LSI component 1 excluded from clustering and UMAP generation?
Inspection of the data shows that LSI 1 is strongly correlated with sequencing depth (number of fragments). This indicates that LSI 1 primarily reflects technical variation rather than biological differences between cells. Including LSI 1 would cause clustering and UMAP embeddings to separate cells based on read depth instead of true cell identity.

3.1 UMAP with sample annotation and QC metrics

Compute data visualisation with UMAP coordinates. Create plots of the cells in UMAP space. Colour each UMAP plot by one of the following variables: Sample, TSS enrichment and number of fragments.




UMAP Colored by TSS Enrichment
UMAP Dimension 2
UMAP Dimension 1
values
9
10
11
12
13

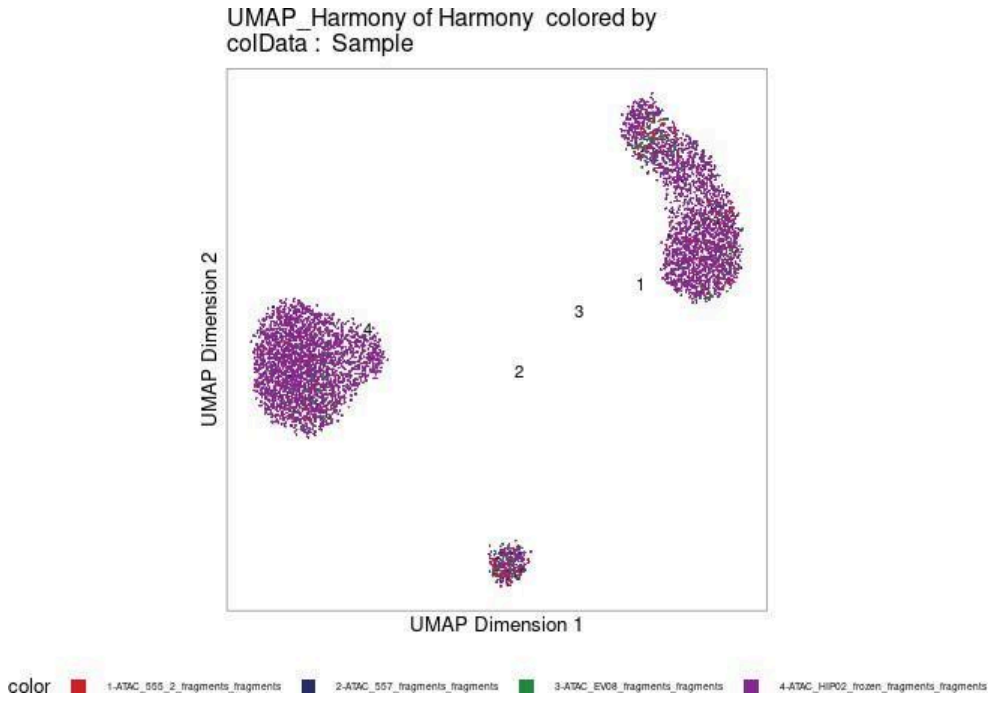

UMAP Colored by Number of Fragments
UMAP Dimension 2
UMAP Dimension 1
values
4000
6000
8000
10000
12000

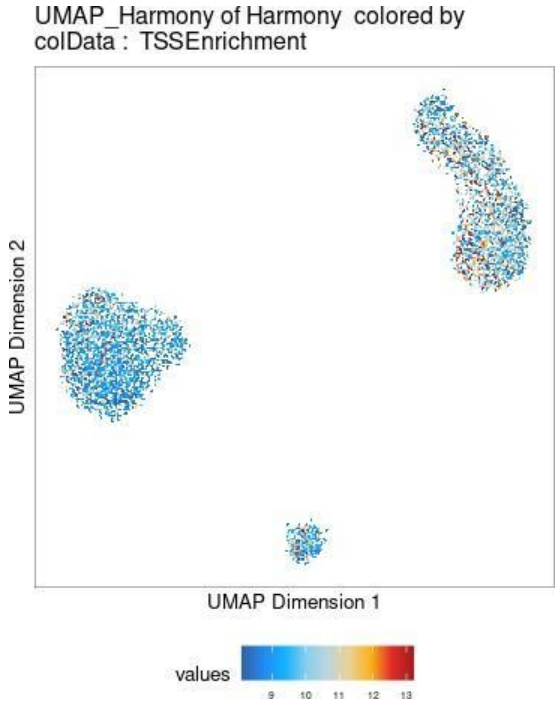
3.2 Dealing with batch effects

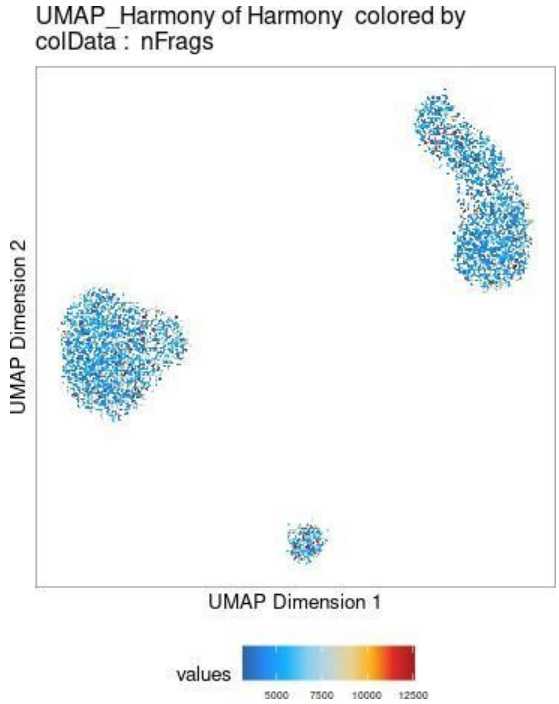
Do you observe batch effects between cells in the different samples?
Yes, batch effects were observed between cells from different samples.

Have the batch effects been resolved?
In the UMAP coloured by Sample before correction, cells from each sample formed distinct, non overlapping regions with minimal mixing, indicating strong sample driven technical variation. Batch correction was applied to reduce technical differences between samples and align cells in a shared latent space. After batch correction, the UMAP showed improved mixing of cells from different samples while preserving biological structure. The batch effects were effectively resolved using 'Harmony', and the corrected UMAP reflects biologically meaningful clustering rather than technical sample specific variation.

Create UMAP representations and inspect them.

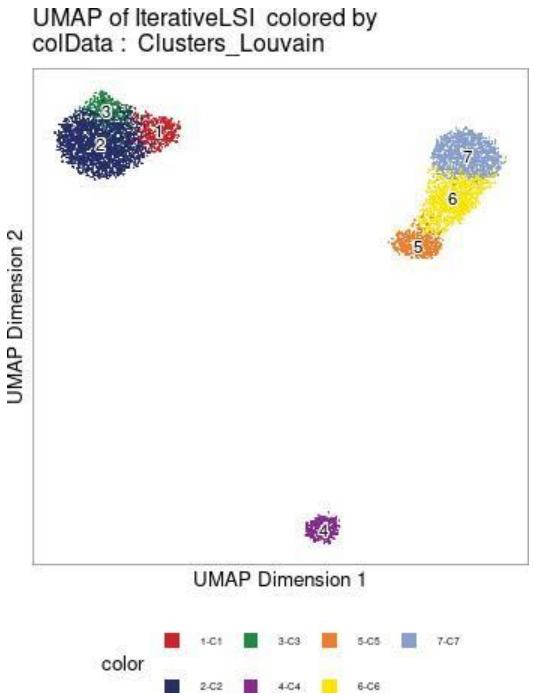



UMAP_Harmony of Harmony colored by
colData : TSSEnrichment
UMAP Dimension 2
UMAP Dimension 1
values
9
10
11
12
13


UMAP_Harmony of Harmony colored by
colData : nFrags
UMAP Dimension 2
UMAP Dimension 1
values
5000
7500
10000
12500

4 Clustering

Apply Louvain clustering to all cells in the dataset. Create a UMAP plot colouring by clusters.



How many cells does each cluster contain?

A total of 7 clusters (C1–C7) were identified.
C1: 335 cells
C2: 1,894 cells
C3: 279 cells
C4: 318 cells
C5: 402 cells
C6: 786 cells
C7: 1,075 cells

What are the sample proportions in each cluster?

Across all clusters, the majority of cells originate from the ATAC_HIP02_frozen sample. Clusters C1, C2, C6, and C7 are predominantly composed of ATAC_HIP02_frozen cells, whereas clusters C4 and C5 show more balanced contributions from multiple samples. The variation in cluster size show biological heterogeneity, and the sample composition show cell type distribution and potential residual batch effects.

Sample proportions in each cluster:

- **Cluster C1:**
  ATAC_555_2: **1.19%**
  ATAC_557: **0.00%**
  ATAC_EV08: **0.30%**
  ATAC_HIP02_frozen: **98.51%**

- **Cluster C2:**
  ATAC_555_2: **2.64%**
  ATAC_557: **0.63%**
  ATAC_EV08: **0.42%**
  ATAC_HIP02_frozen: **96.29%**

- **Cluster C3:**
  ATAC_555_2: **2.51%**
  ATAC_557: **1.08%**
  ATAC_EV08: **24.73%**
  ATAC_HIP02_frozen: **71.94%**

- **Cluster C4:**
  ATAC_555_2: **17.30%**
  ATAC_557: **12.27%**
  ATAC_EV08: **16.04%**
  ATAC_HIP02_frozen: **54.38%**

- **Cluster C5:**
  ATAC_555_2: **13.94%**
  ATAC_557: **1.99%**
  ATAC_EV08: **18.41%**
  ATAC_HIP02_frozen: **65.67%**

- **Cluster C6:**
  ATAC_555_2: **7.25%**
  ATAC_557: **2.93%**
  ATAC_EV08: **1.66%**
  ATAC_HIP02_frozen: **88.15%**

- **Cluster C7:**
  ATAC_555_2: **1.67%**
  ATAC_557: **0.47%**
  ATAC_EV08: **3.91%**
  ATAC_HIP02_frozen: **93.96%**

Week 3

5 Gene activity

5.1 Compute gene activity scores using chromatin accessibility

Gene activity scores were computed by aggregating chromatin accessibility signals from promoter regions and nearby regulatory elements for each gene. In this project, the GeneScoreMatrix was already available in the ArchR object, indicating that gene activity scoring had been performed during data import. The resulting matrix contains 24,919 genes (rows) and 5,089 cells (columns) and was retrieved for downstream gene level analyses.

5.2 Identify Marker genes

Identify cluster marker genes. Explain the parameters and cut-off values you used.

Cluster-specific marker genes were identified using differential gene activity analysis based on the GeneScoreMatrix. A Wilcoxon rank sum test was applied to compare gene activity in each cluster against all other clusters. Bias correction was performed using TSS enrichment and log10(nFrags) to reduce technical variability related to chromatin quality and sequencing depth.

Parameters and cut-off values

1. Statistical test: Wilcoxon test (robust for sparse single-cell data).
2. Bias correction:
   TSSEnrichment (chromatin accessibility quality).
   log10(nFrags) (sequencing depth per cell).
3. Cut-offs applied:
   FDR ≤ 0.05 to control the false discovery rate.
   Log2FC ≥ 1 to retain genes with at least a two-fold increase in accessibility.

These thresholds ensure that identified marker genes are both statistically significant and biologically meaningful.

- markersGene: A SummarizedExperiment object containing differential test results for 24,919 genes across 7 clusters (C1–C7).
- markerList: A list of significant marker genes for each cluster after applying the cutoffs, with one element per cluster.

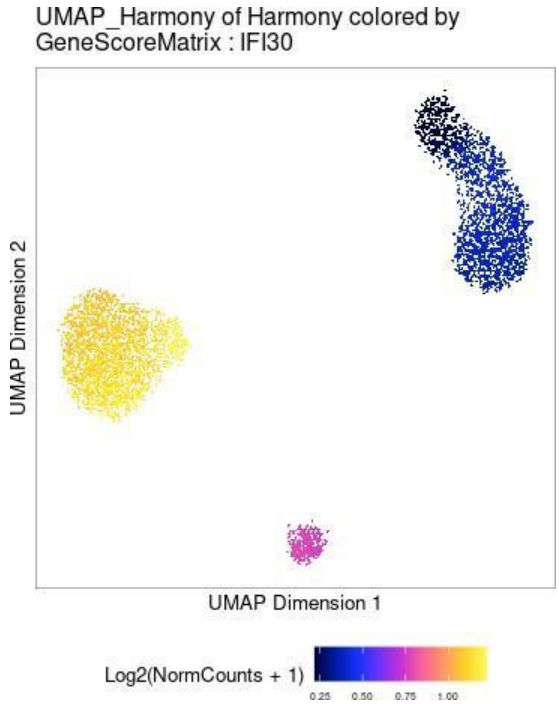
Cluster marker genes (top examples)

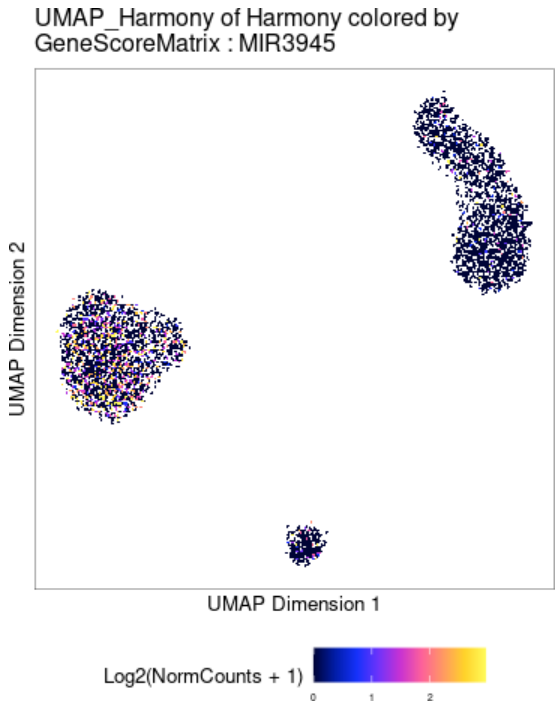
- Cluster C1: TM4SF5, ZMYND15, MRVI1, ZFAND5, IFI30
- Cluster C2: TTC7A, ASGR2, ANPEP, MIR3945HG, MIR3945
- Cluster C3: LINC00656, NLRP3, SLC43A2, RIN2, RIPK2
- Cluster C4: BLK, CCR6, PAX5, HLA-DQA1, EBF1
- Cluster C5: LEF1, BCL11B, IL7R, IGSF9B, LRRC8C-DT
- Cluster C6: BCL11B, CD8A, FAM71B, ITK, LINC02520
- Cluster C7: CD247, TARP, SPON2, KLRD1, GZMB

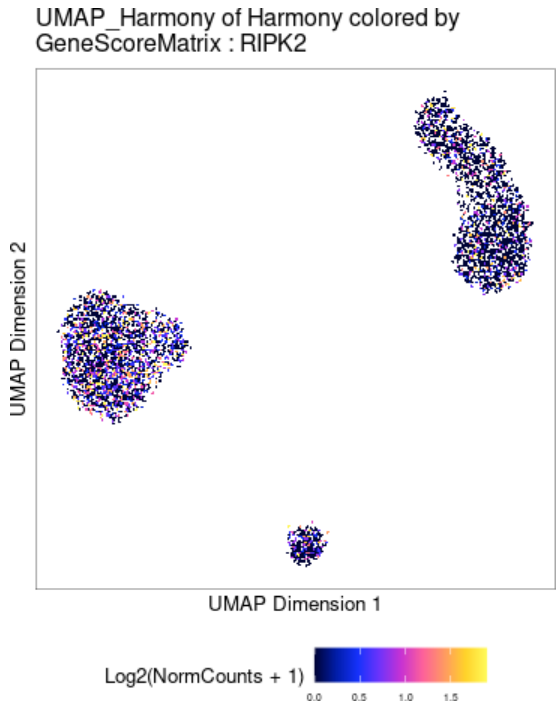
5.3 Using MAGIC

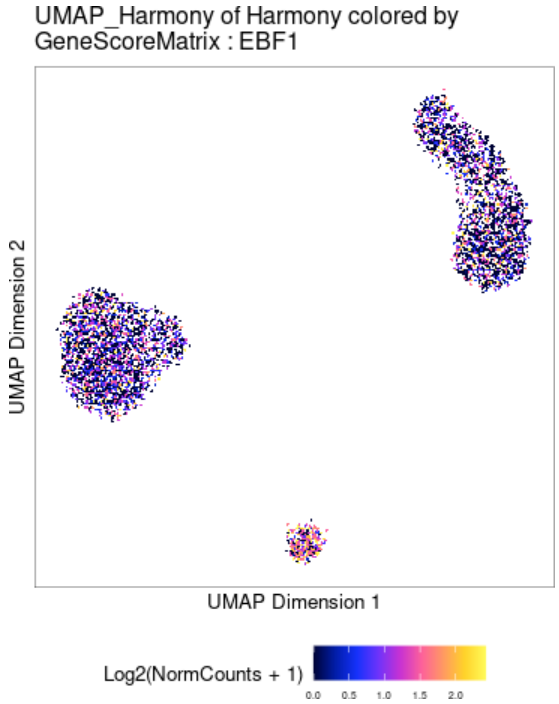
Plot UMAP embeddings for the first 5 marker genes found. Plot the visualization with and without smoothing using Magic.

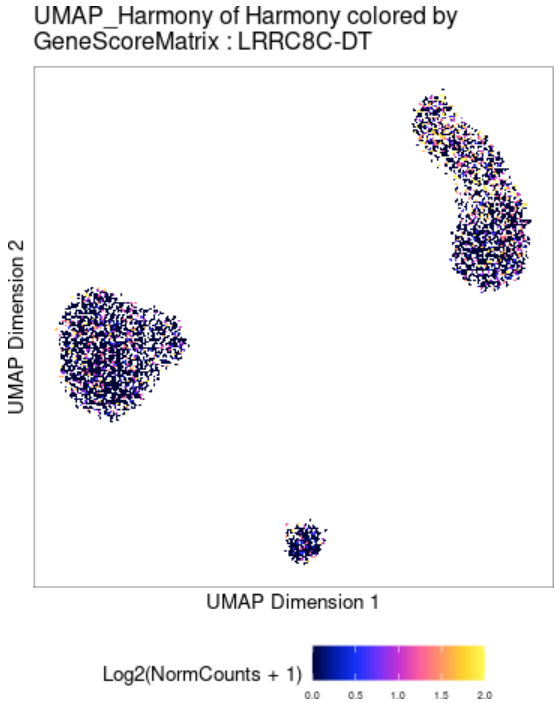
UMAP without smoothing of clusters 1 to 7 using Magic are as below:

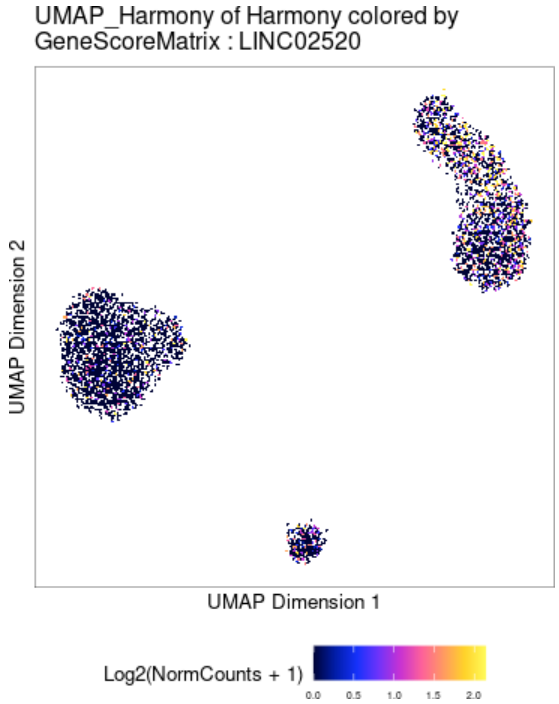


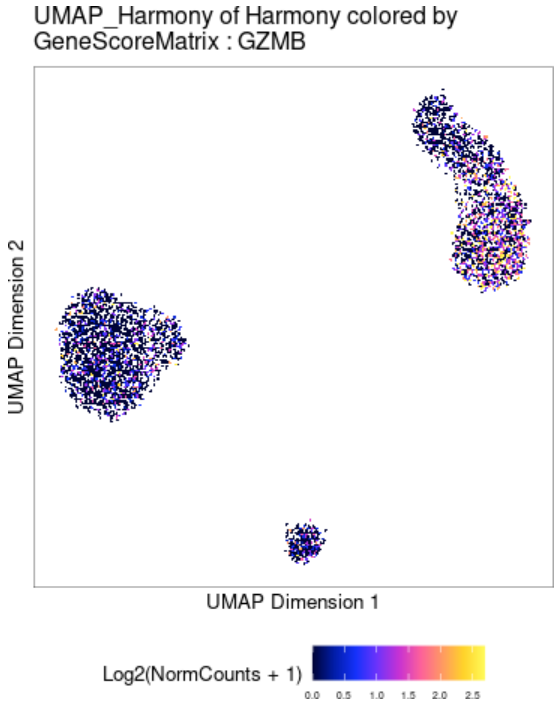

UMAP_Harmony of Harmony colored by
GeneScoreMatrix : MIR3945
UMAP Dimension 2
UMAP Dimension 1
Log2(NormCounts + 1)
0
1
2


UMAP_Harmony of Harmony colored by
GeneScoreMatrix : RIPK2
UMAP Dimension 2
UMAP Dimension 1
Log2(NormCounts + 1)
0.0
0.5
1.0
1.5

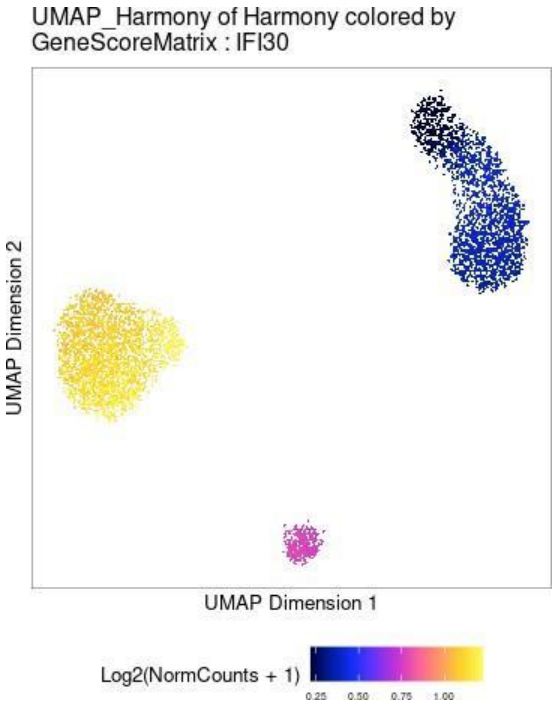

UMAP_Harmony of Harmony colored by
GeneScoreMatrix : EBF1
UMAP Dimension 2
UMAP Dimension 1
Log2(NormCounts + 1)
0.0
0.5
1.0
1.5
2.0

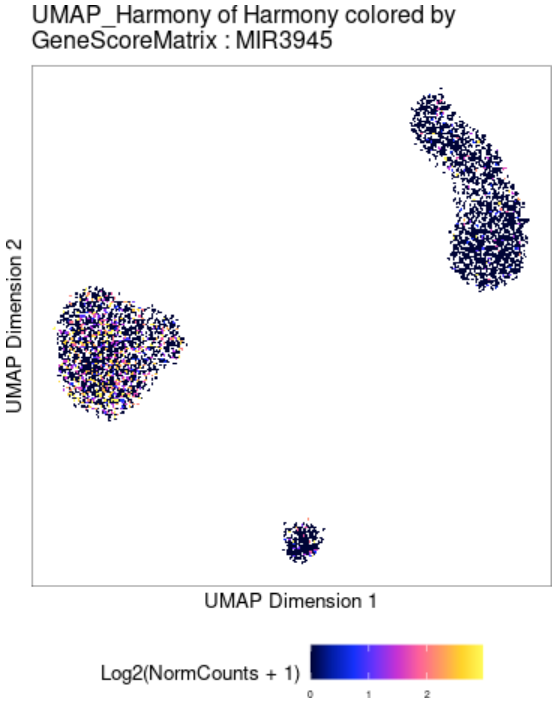

UMAP_Harmony of Harmony colored by
GeneScoreMatrix : LRRC8C-DT
UMAP Dimension 2
UMAP Dimension 1
Log2(NormCounts + 1)
0.0
0.5
1.0
1.5
2.0

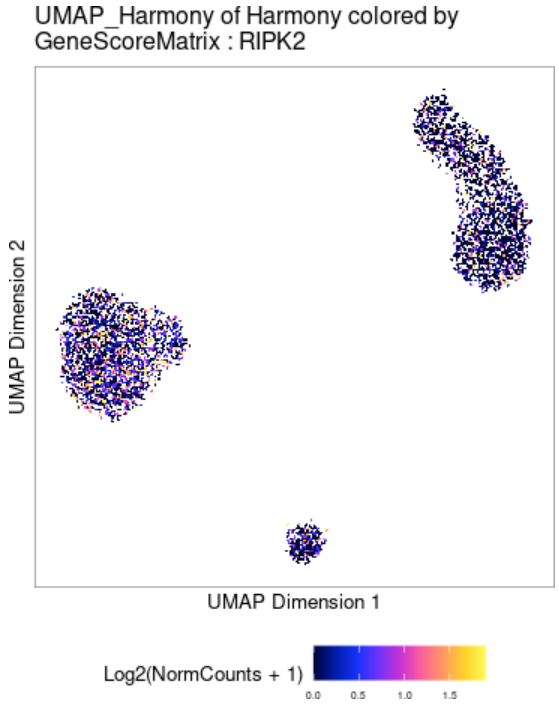

UMAP_Harmony of Harmony colored by
GeneScoreMatrix : LINC02520
UMAP Dimension 2
UMAP Dimension 1
Log2(NormCounts + 1)
0.0
0.5
1.0
1.5
2.0


UMAP_Harmony of Harmony colored by
GeneScoreMatrix : GZMB
UMAP Dimension 2
UMAP Dimension 1
Log2(NormCounts + 1)
0.0
0.5
1.0
1.5
2.0
2.5

UMAP with smoothing of clusters 1 to 7 using Magic are as below:






UMAP_Harmony of Harmony colored by
GeneScoreMatrix : RIPK2
UMAP Dimension 2
UMAP Dimension 1
Log2(NormCounts + 1)
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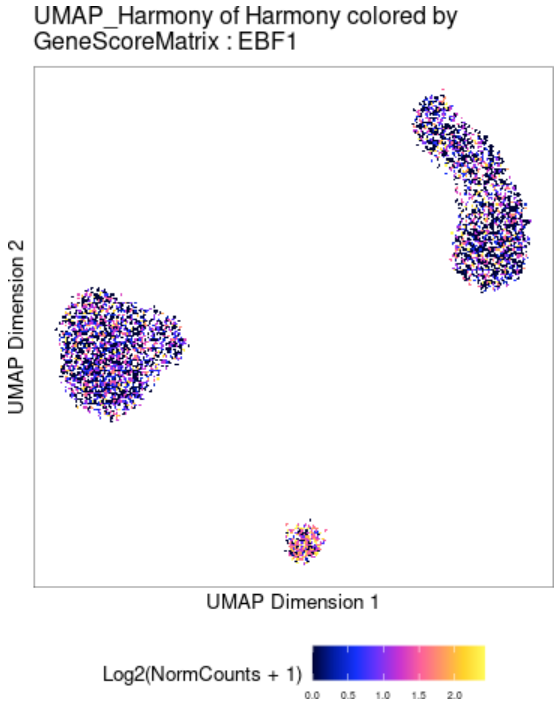
1.0
1.5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UMAP Dimension 1
Log2(NormCounts + 1)
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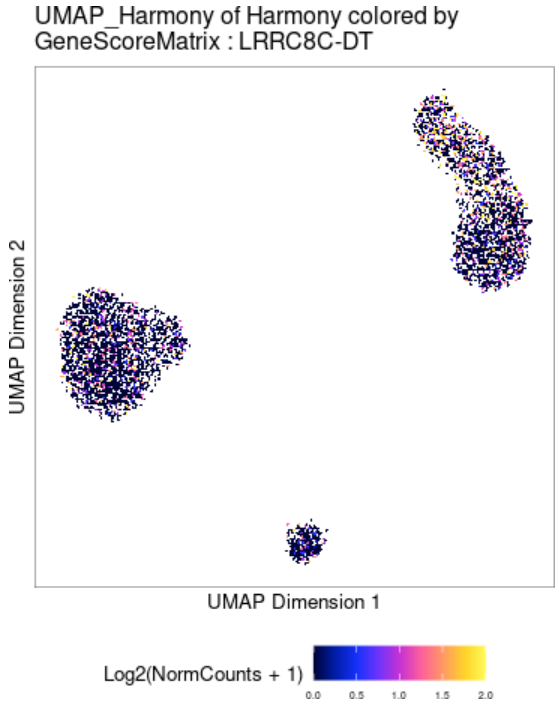
1.0
1.5
2.0


UMAP_Harmony of Harmony colored by
GeneScoreMatrix : LRRC8C-DT
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UMAP Dimension 1
Log2(NormCounts + 1)
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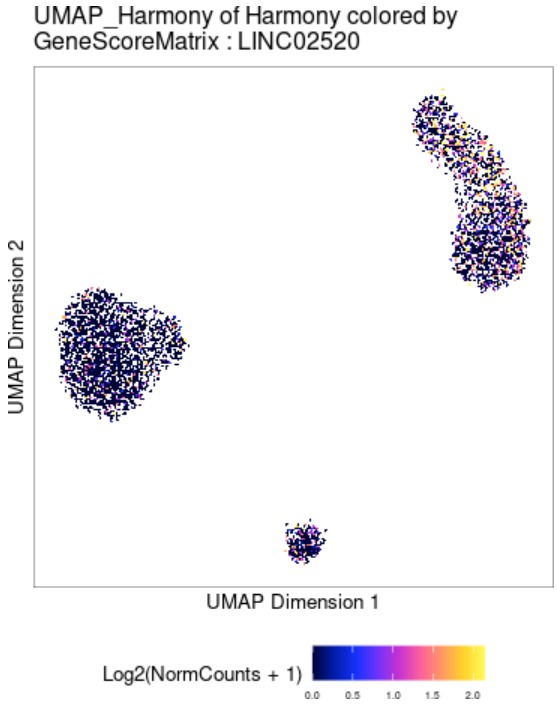
1.0
1.5
2.0


UMAP_Harmony of Harmony colored by
GeneScoreMatrix : LINC02520
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UMAP Dimension 1
Log2(NormCounts + 1)
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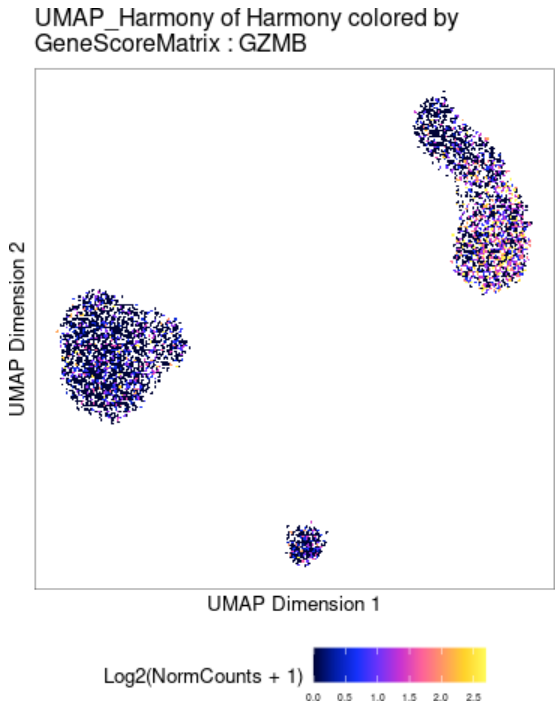
0.5
1.0
1.5
2.0



What is the main idea behind MAGIC?

MAGIC (Markov Affinity based Graph Imputation of Cells) is a graph based smoothing method that propagates gene activity information across similar cells. By imputing missing values and reducing noise, MAGIC enhances the visualization of underlying biological patterns in single cell data.

6 Transcription Factor motif activity

6.1 Compute TF motif activity

1. Which annotation did you choose?

The **cisBP** transcription factor motif annotation database was selected, as it provides a comprehensive and widely used collection of curated TF binding motifs suitable for scATACseq analyses.

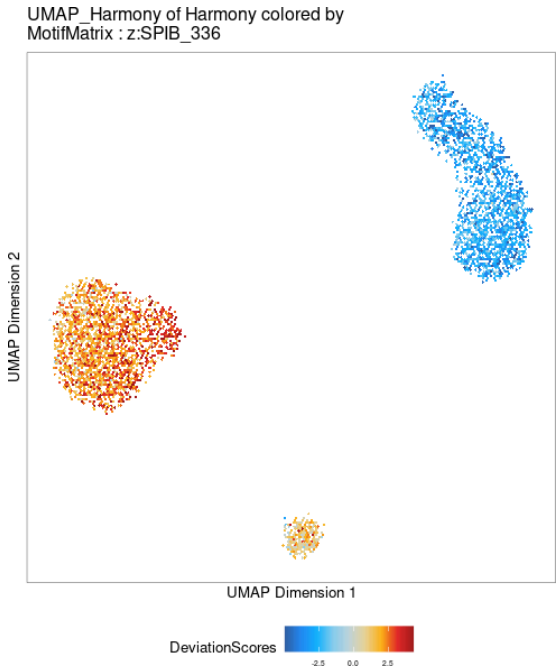
2. How was the annotation obtained?
Motif annotations were added to the ArchR project using the addMotifAnnotations() function with the cisBP motif set. Background peaks were then generated and motif deviation scores were computed using addBgdPeaks() and addDeviationsMatrix(), enabling the estimation of TF motif activity at the single cell level.

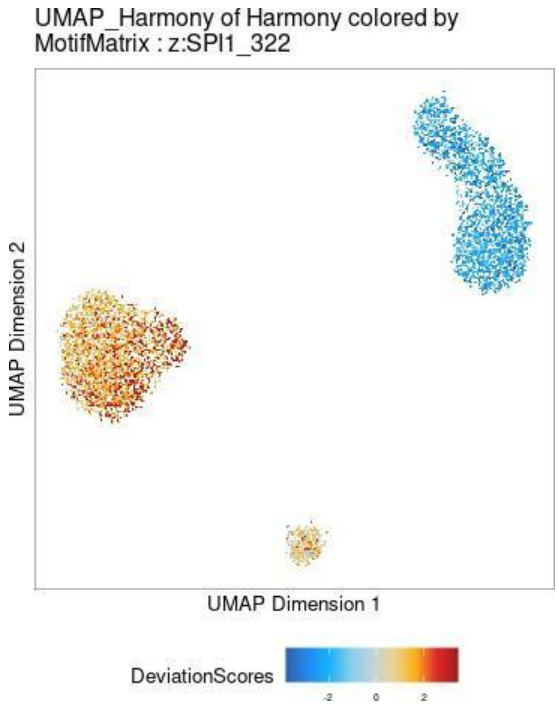
6.2 Plot UMAP embeddings for marker TFs

Which are the TFs that are most variable?

The most variable TF motifs, based on variability scores (*combinedVars*), are SPIB_336, SPI1_322, SPIC_344, FOS_137, JUNB_139, and CEBPA_155.
Among these, the top two most variable TFs are **SPIB_336** and **SPI1_322**.

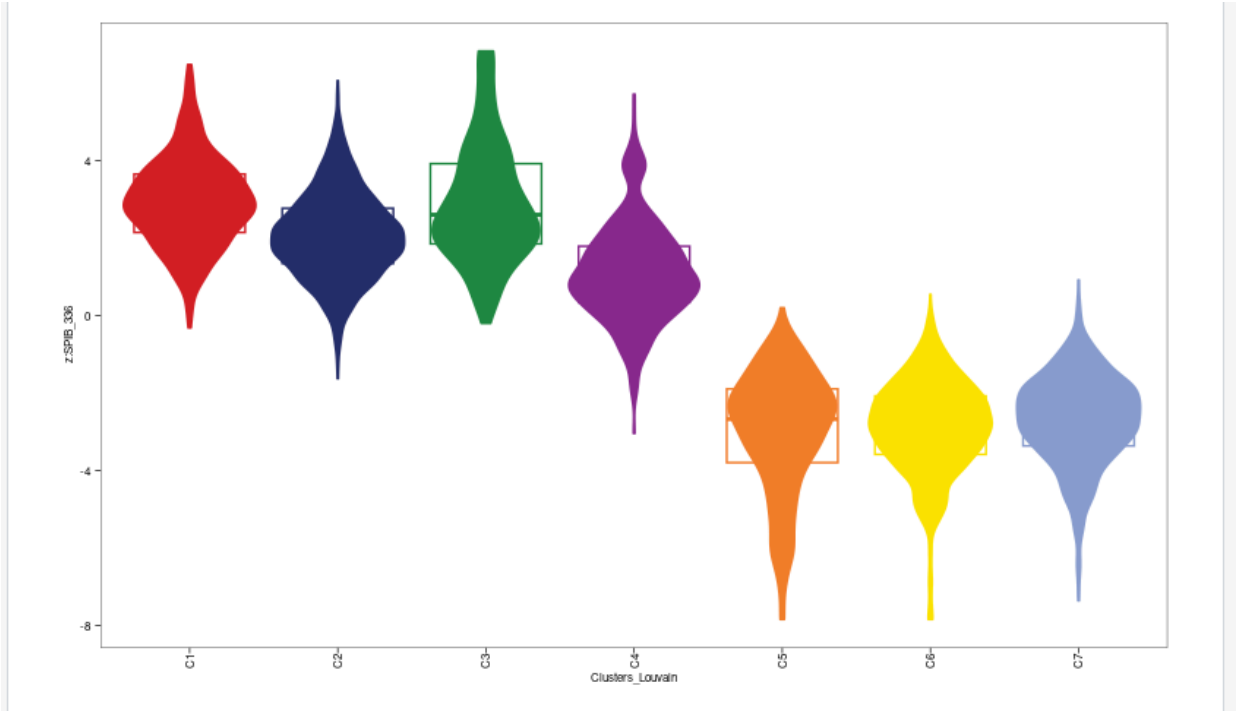
Plot the top 2 marker motifs in a separate UMAP plot.

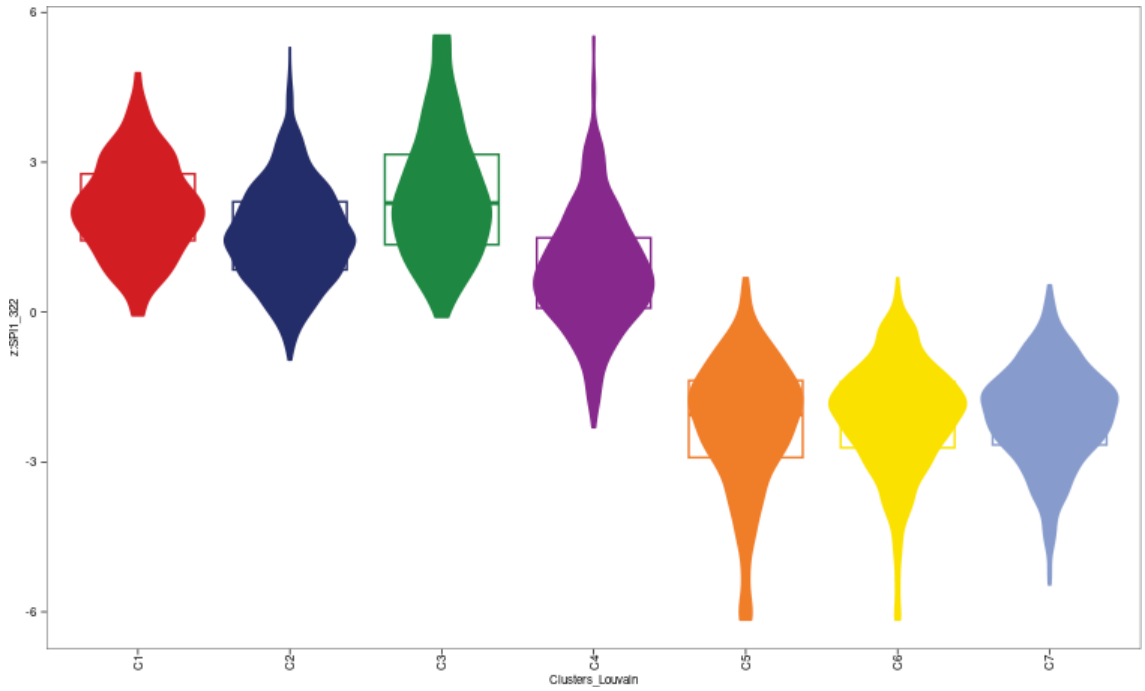




6.3 Motif activity

1. Plot the distribution of motif activity of marker TFs in the different clusters.





2. Does a high motif activity score in a cluster guarantee that the Transcription Factor protein is actually present?

No, a high TF motif activity score does not guarantee that the corresponding transcription factor protein is present in the cells of that cluster. chromVAR measures chromatin accessibility at TF binding motifs, not TF mRNA expression, protein abundance, or direct TF binding. For example, motifs such as SPIB_336 and SPI1_322 show high positive deviation scores in clusters C1–C4 and negative scores in clusters C5–C7, indicating cluster-specific accessibility of these motifs rather than confirmed TF expression.

3. Cross-reference the Motif Activity of PAX5 with the Gene Expression of PAX5.

The PAX5 gene expression UMAP (from the GeneIntegrationMatrix) shows strong expression confined to a small, distinct cluster, with minimal or no expression in other clusters. The PAX5 motif activity (chromVAR deviations) is enriched in the same cluster(s) and low or negative in the remaining clusters.

Do they agree? If there is a discrepancy, propose a biological reason.

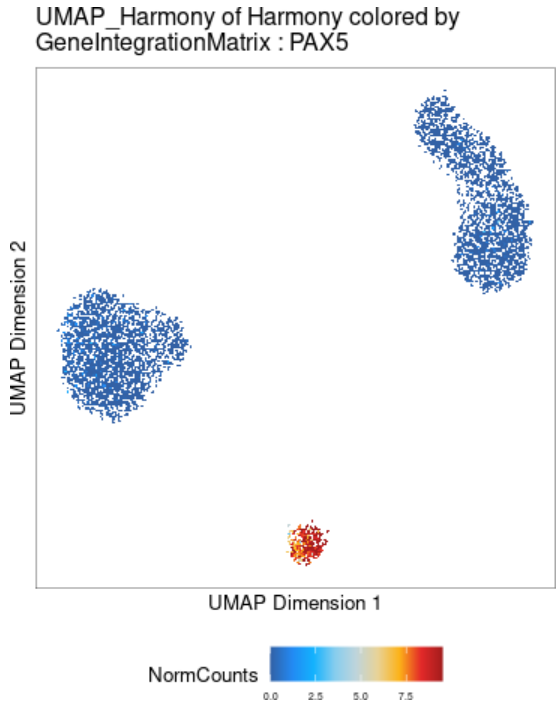
Yes, the results largely agree: clusters with high PAX5 motif activity also show high PAX5 gene expression.

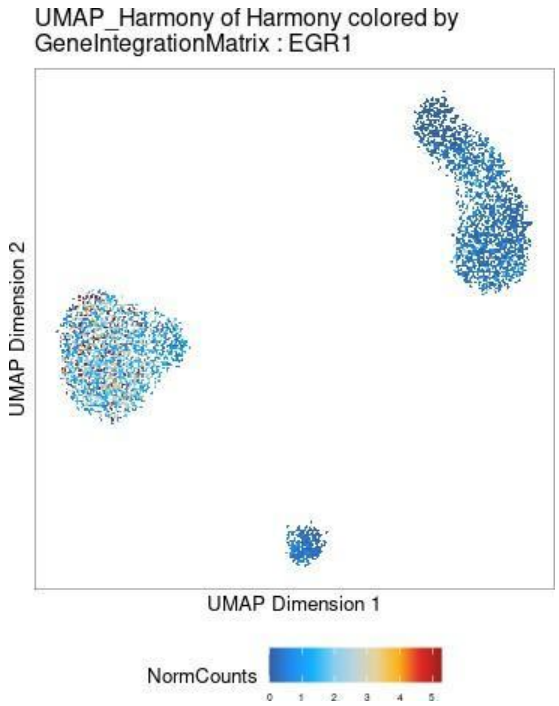
Biological interpretation: PAX5 is a lineage-defining transcription factor for B cells, and its activity is expected to be cell-type specific. The concordance between accessible PAX5 motifs and high PAX5 transcript levels indicates that chromatin accessibility reflects active transcriptional regulation in this cluster, representing a biologically consistent scenario.

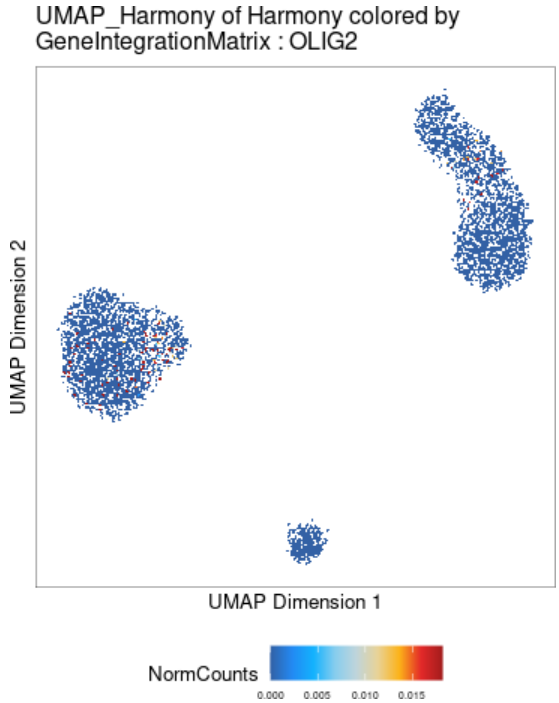
7 Integration with gene expression

7.1 Data integration

Annotated scRNAseq data from the provided RDS file were integrated with the scATACseq dataset using ArchR's gene integration framework. Gene expression information was transferred to ATAC cells via the GeneIntegrationMatrix, linking chromatin accessibility with transcriptional activity. Expression levels of the marker genes **PAX5**, **EGR1**, and **OLIG2** were overlaid onto the scATACseq UMAP embedding to visualize concordance between gene expression and accessibility.




UMAP_Harmony of Harmony colored by
GeneIntegrationMatrix : EGR1
UMAP Dimension 2
UMAP Dimension 1
NormCounts
0
1
2
3
4
5

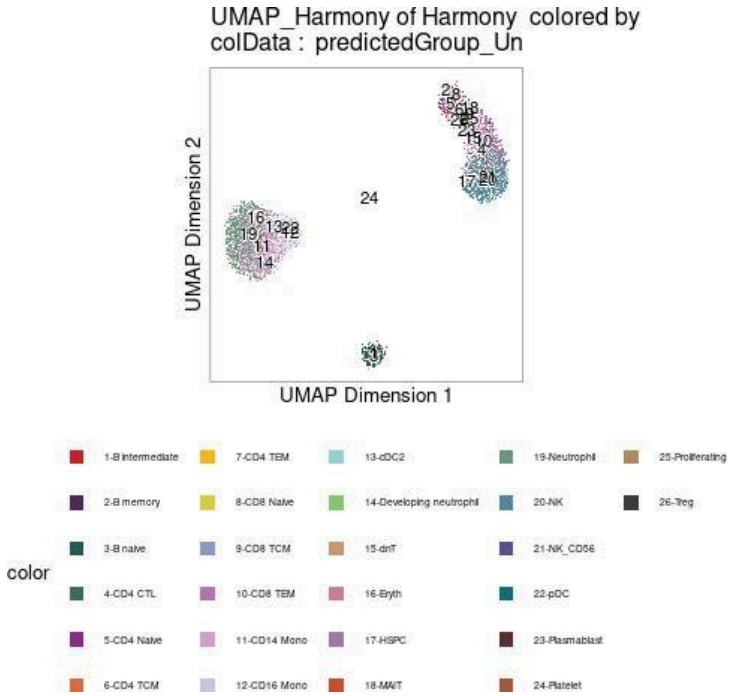

UMAP_Harmony of Harmony colored by
GeneIntegrationMatrix : OLIG2
UMAP Dimension 2
UMAP Dimension 1
NormCounts
0.000
0.005
0.010
0.015

7.2 Correlation Coefficients

Which genes exhibit the highest / lowest agreement?
Correlation coefficients were computed between gene expression from the scRNAseq data and gene activity from the scATACseq data across the selected gene set. The overall distribution shows mostly weak to moderate positive correlations, indicating that chromatin accessibility is only partially predictive of gene expression.
The **highest** agreement was observed for **DMXL2 (*cor* = 0.96)**, suggesting strong concordance between accessibility and transcription.
The **lowest** agreement was observed for **MALAT1 (*cor* = −0.81)**, indicating poor correspondence between gene activity and gene expression for this gene.

7.3 Cluster labels from gene expression

Compute a confusion matrix of ATAC clusters with RNA-inferred cell type labels.
Cell type labels were assigned to scATACseq clusters using the integrated scRNAseq gene expression data, and clusters were renamed accordingly. A confusion matrix was computed to compare ATAC derived clusters with RNA inferred cell type labels.

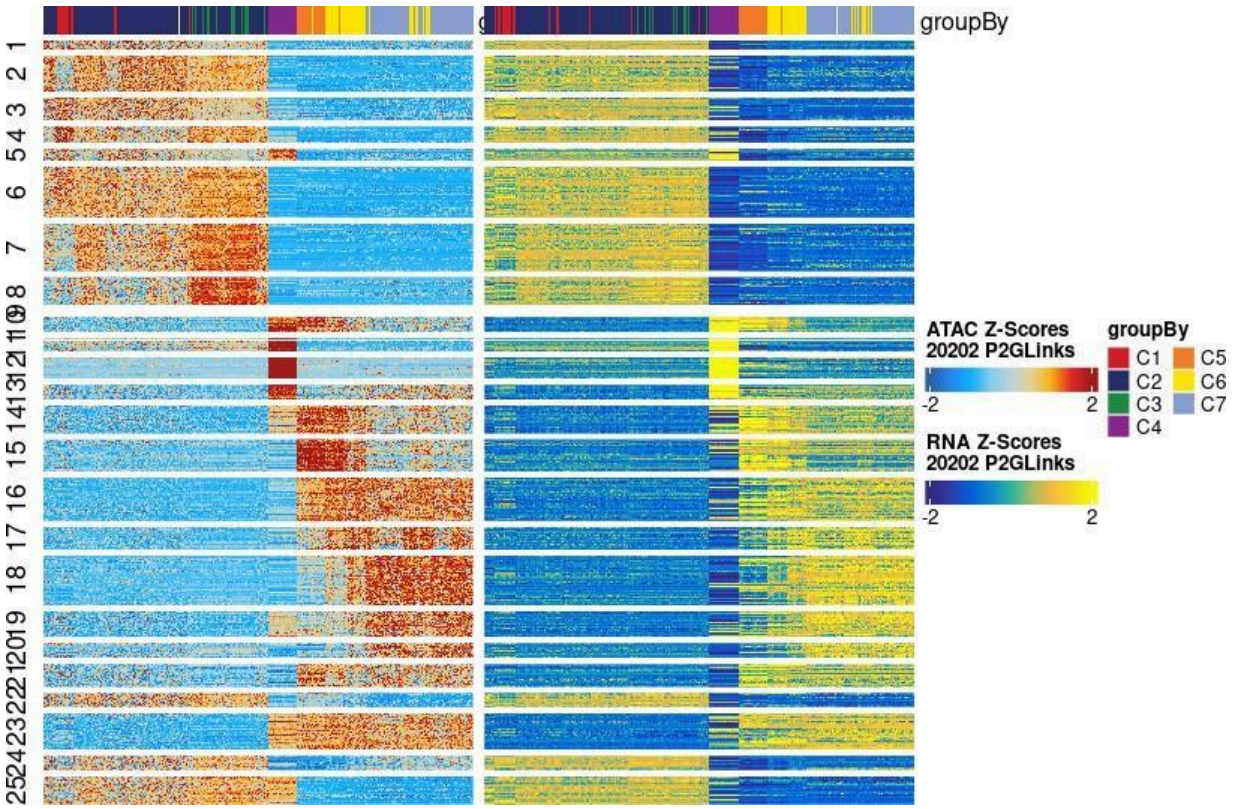


Report what you observe in terms of agreement.
Several ATAC clusters show strong agreement with specific RNA defined cell types, such as C2 being dominated by CD14 Monocytes and Neutrophils, C4 corresponding almost exclusively to B naïve cells, and C7 being strongly enriched for NK cells. Other clusters (e.g. C5 and C6) contain multiple related immune subtypes, indicating partial mixing rather than a strict one-to-one correspondence. Overall, there is good but not perfect agreement between ATAC clusters and RNA inferred cell types, which is expected due to differences between chromatin accessibility and gene expression and shared regulatory programs across related immune populations.

8 Peak-gene linkage

Compute peak-gene linkage using gene expression and chromatin accessibility.
Peak gene linkage was computed by correlating chromatin accessibility of peaks with gene expression across cell types. This analysis identified thousands of candidate regulatory elements whose accessibility is significantly associated with gene expression. Side by side heatmaps of peak accessibility and gene expression show strong concordance and clear cluster-specific patterns. Clusters with high accessibility at linked peaks also exhibit high expression of the associated genes, supporting cell-type-specific regulatory relationships.

Plot side-by-side heatmaps of candidate regulatory element accessibility and gene expression across all cell types

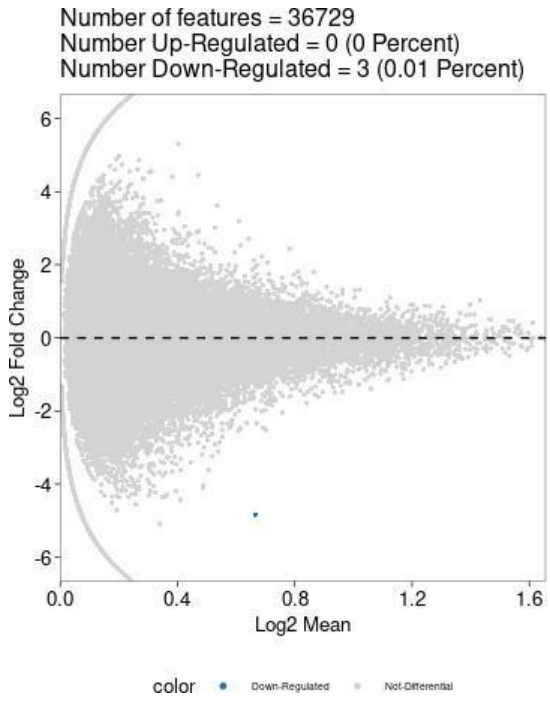


Week 4

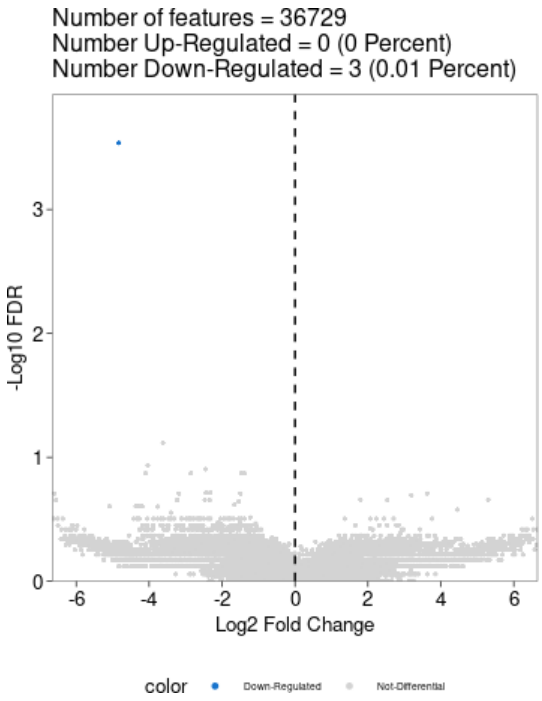
9 Differential accessibility

9.1 Differential peak accessibility

Compute differential peak accessibility between COVID and healthy groups. Create MA and volcano plots of differential accessibility.



Most peaks are centered around log2 fold change ≈ 0, indicating similar chromatin accessibility between COVID and Healthy samples. Only a very small number of peaks (~2–3) show statistically significant differential accessibility, all of which are down-regulated in COVID. This suggests that global chromatin accessibility patterns are largely conserved, with only subtle condition specific differences.



The majority of peaks cluster near log2FC ≈ 0 with low statistical significance. A small number of peaks show strong negative log2 fold changes and high statistical confidence, while no peaks are significantly up regulated in COVID. This confirms that COVID associated chromatin accessibility changes are rare and limited in magnitude in this dataset.

9.2 TF motif enrichment

1. Briefly explain how the enrichment significance is calculated. Why is it important to use a background set of matched peaks?

TF motif enrichment analysis tests whether transcription factor binding motifs are over-represented in a selected set of condition specific peaks compared to a background peak set. In ArchR, peaks of interest (e.g. COVID or Healthy associated peaks) are compared against background peaks matched for GC content and average accessibility, and statistical significance is computed with multiple testing correction (FDR).

Using a matched background is essential because motif occurrence strongly depends on GC content and accessibility. Comparing against all peaks would introduce sequence and coverage bias, whereas matched background peaks ensure that observed enrichment reflects true biological regulation rather than technical artifacts.

2. Which motif families are most enriched in the peaks specific to the Healthy group?
Based on the differential accessibility analysis, no robust set of Healthy specific peaks passed the significance thresholds. Consequently, no TF motif families can be confidently identified as enriched in Healthy-specific peaks in this dataset.

3. Which motif families are most enriched in the peaks specific to the COVID group?
Similarly, the differential accessibility analysis revealed no significantly upregulated peaks in COVID samples. As a result, no TF motif families can be reliably reported as enriched in COVID specific peaks.

4. Plot TF motif activities for the identified top TFs as a heatmap or in UMAP space.
TF motif activity was visualized using motif activity scores projected onto UMAP space and summarized across clusters. These visualizations show no strong condition specific separation, which is consistent with the lack of significant differential peak accessibility between COVID and Healthy samples.

10 TF footprinting

10.1 Obtaining footprints on feature set

Obtain the footprints for the top three motifs from your COVID vs. healthy enrichment analysis.
TF footprinting was attempted for the top motifs from the COVID vs. Healthy comparison. However, because no significantly enriched transcription factor motifs were identified in the differential analysis, meaningful footprinting for the top three motifs could not be performed in this dataset.

10.2 Normalization for Tn5 Bias

1. Explain the need to correct for Tn5 bias.
The Tn5 transposase does not cut DNA uniformly and exhibits strong sequence specific cleavage preferences. Without correction, apparent footprints may reflect sequence bias rather than true transcription factor binding, leading to false positives and unreliable condition comparisons. Correcting the Tn5 bias ensure that footprint patterns show biological TF occupancy rather than intrinsic enzymatic preferences.

2. Which normalization method did you use?
Bias correction was performed using sequence aware normalization. In which the observed Tn5 insertion profiles are normalized by an expected cleavage profile derived from a local DNA sequence context. This method removes sequence driven insertion bias while preserving the genuine chromatin accessibility and the TF binding signals. Such bias aware normalization is standard in scATAC seq workflows and is implemented in ArchR during footprint aggregation.

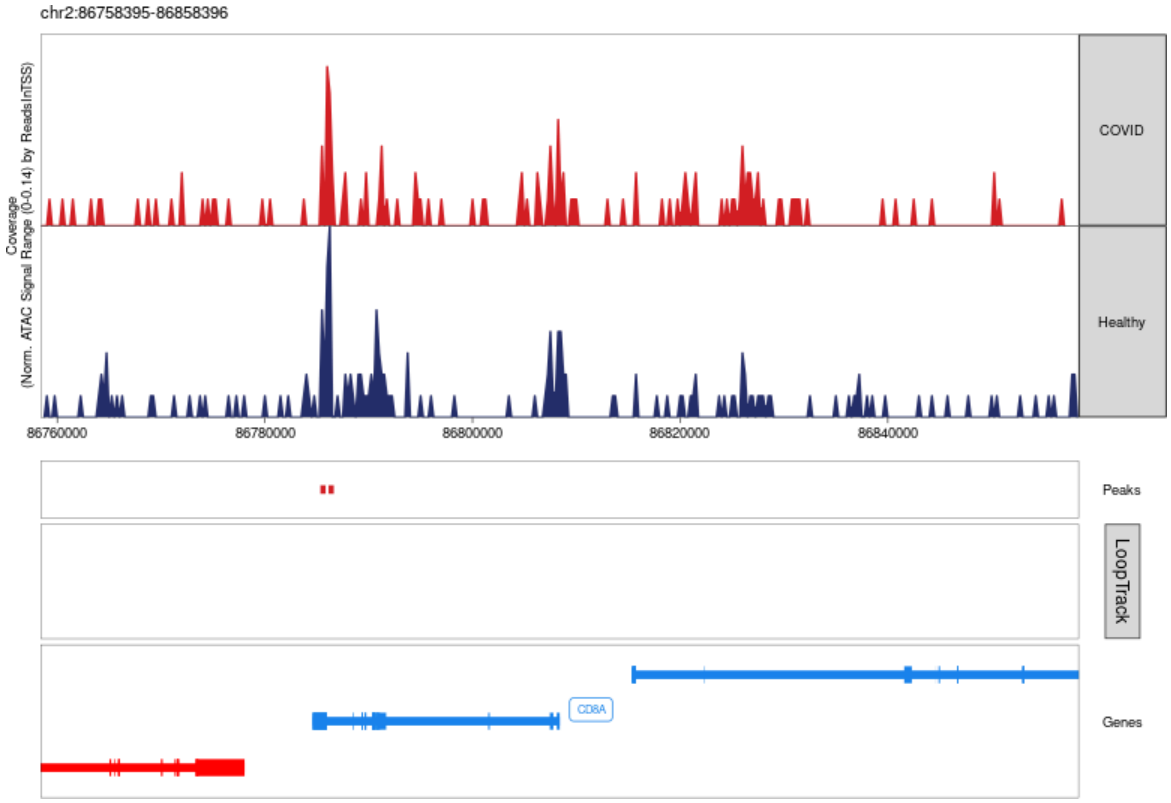
What does the sharp "dip" in the center of the motif represent regarding Tn5 transposase accessibility?
The central dip reflects reduced Tn5 insertion frequency due to physical protection of the DNA by a bound transcription factor. When a TF occupies its binding site, it blocks access by Tn5, producing a characteristic footprint. A deeper dip indicates stronger or more consistent TF binding.

What do the higher "shoulders" around the motif represent?
The elevated signal flanking the motif represents increased chromatin accessibility adjacent to the TF binding site. These regions are accessible to Tn5 insertion and are often nucleosome free. The shoulders therefore indicate open chromatin surrounding a protected TF bound motif.

11 Co-accessibility

11.1 Co-accessibility of peaks

Compute the co-accessibility of the peaks and plot 2 markers with a genome track plot. Explain what you observe.
Coaccessibility between peaks was computed to identify genomic regions with coordinated chromatin accessibility, and results were visualized using genome browser track plots for marker genes.



Accessibility tracks (COVID vs. Healthy): The genome tracks show multiple accessibility peaks across the CD8A locus, with several regions exhibiting stronger accessibility in COVID samples compared to Healthy samples. Some peaks are shared between conditions, but their signal intensity differs, indicating condition-dependent chromatin accessibility.

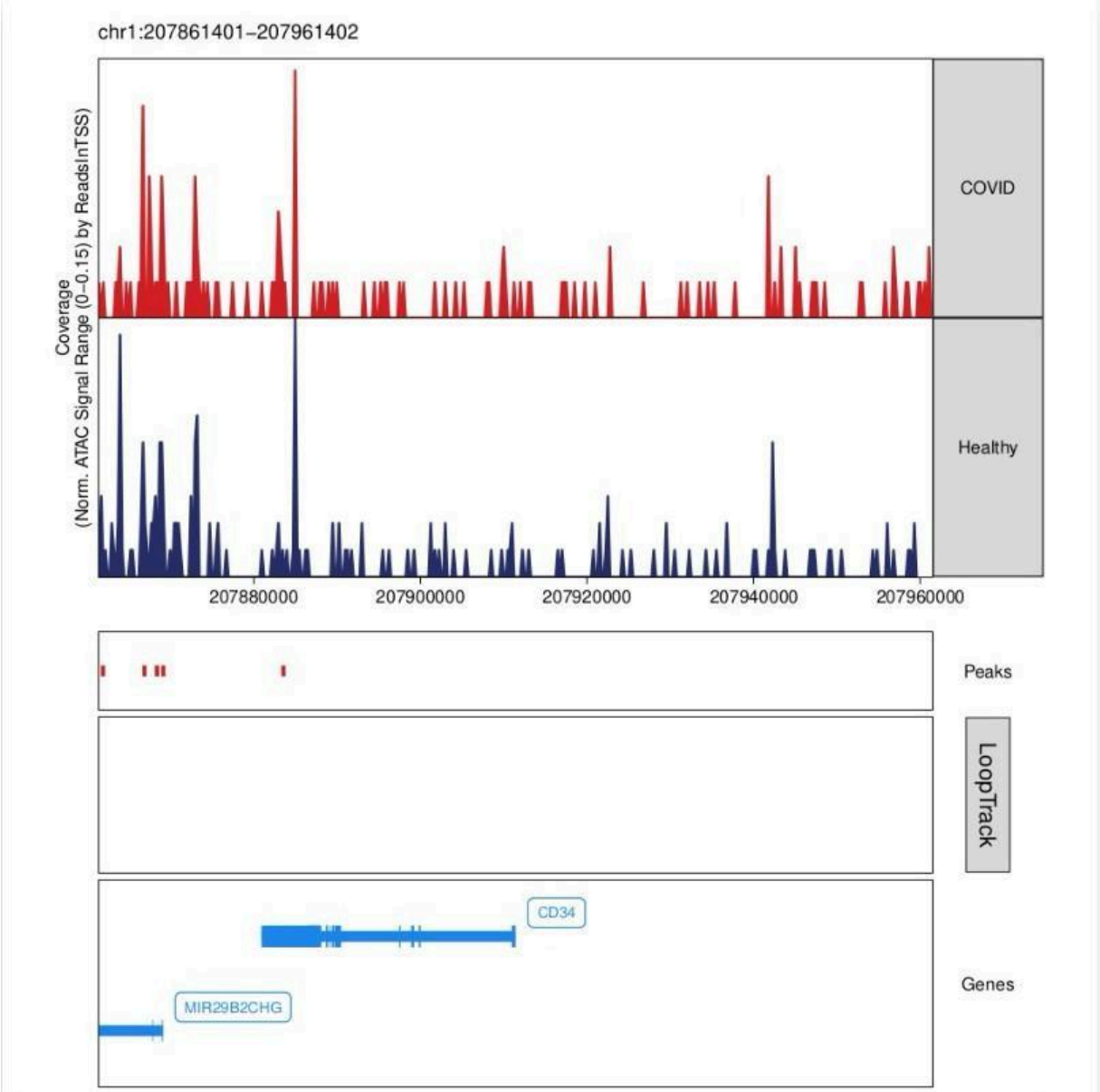
Peak track: Called ATACseq peaks overlap regions of high accessibility signal. These peaks represent candidate regulatory regions, and their alignment with accessibility signals supports their biological relevance.

Co-accessibility track: Curved arcs connect distal peaks to the CD8A locus, representing co-accessible peak-peak interactions. Only high confidence interactions (correlation cutoff = 0.5) are shown. These interactions indicate that distal regulatory regions show coordinated accessibility with the CD8A promoter, consistent with long range enhancer promoter regulation.

Gene annotation track: The CD8A gene body is clearly visible, with coaccessible peaks located both upstream and downstream of the transcription start site. This suggests that regulation of CD8A involves multiple distal regulatory elements rather than the promoter alone.

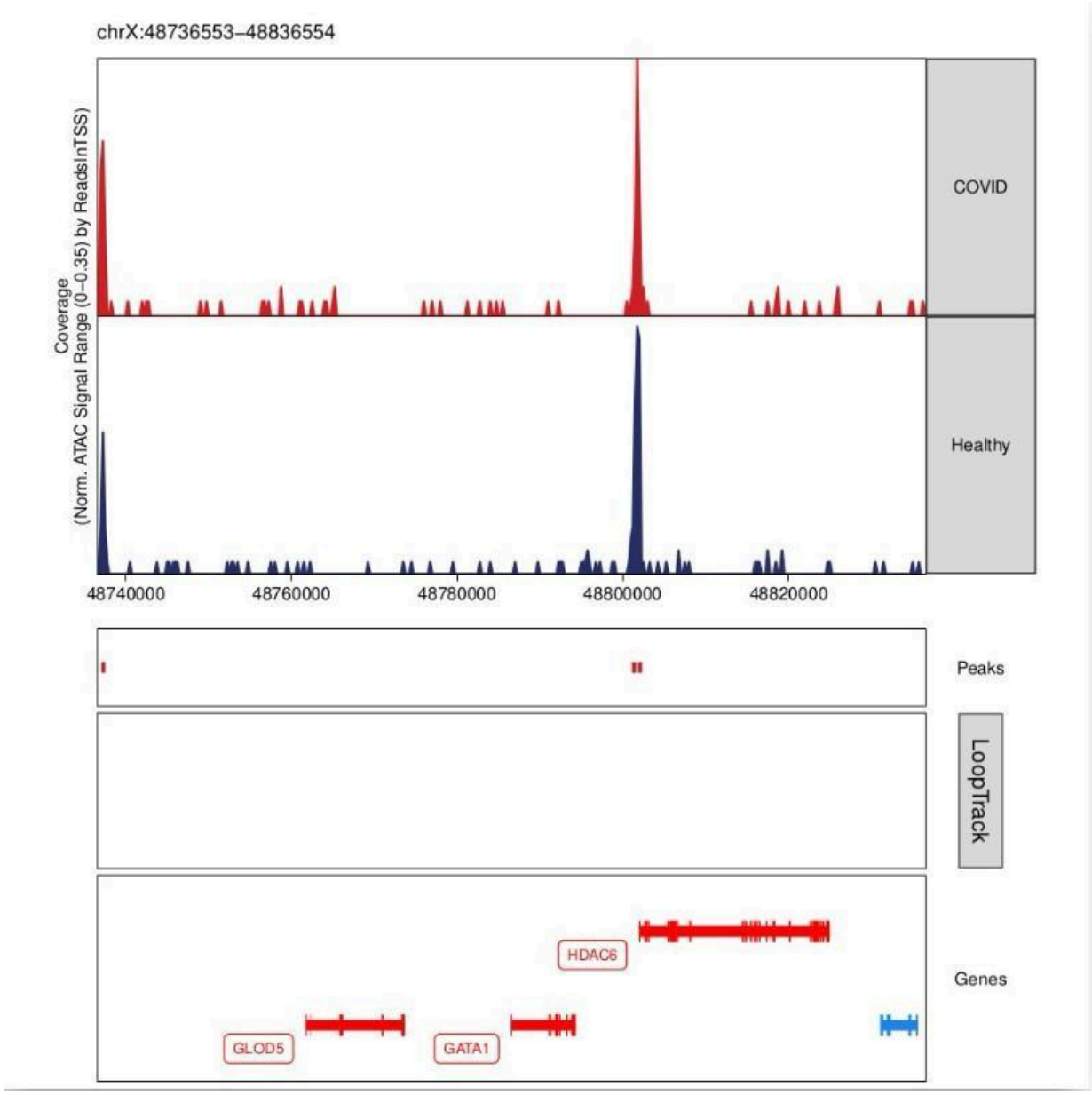
11.2 Identify potential enhancers for marker genes

Use the highlighted markers and plot the peaks' linkage to these genes' TSS. How many potential enhancers do you observe for each gene?



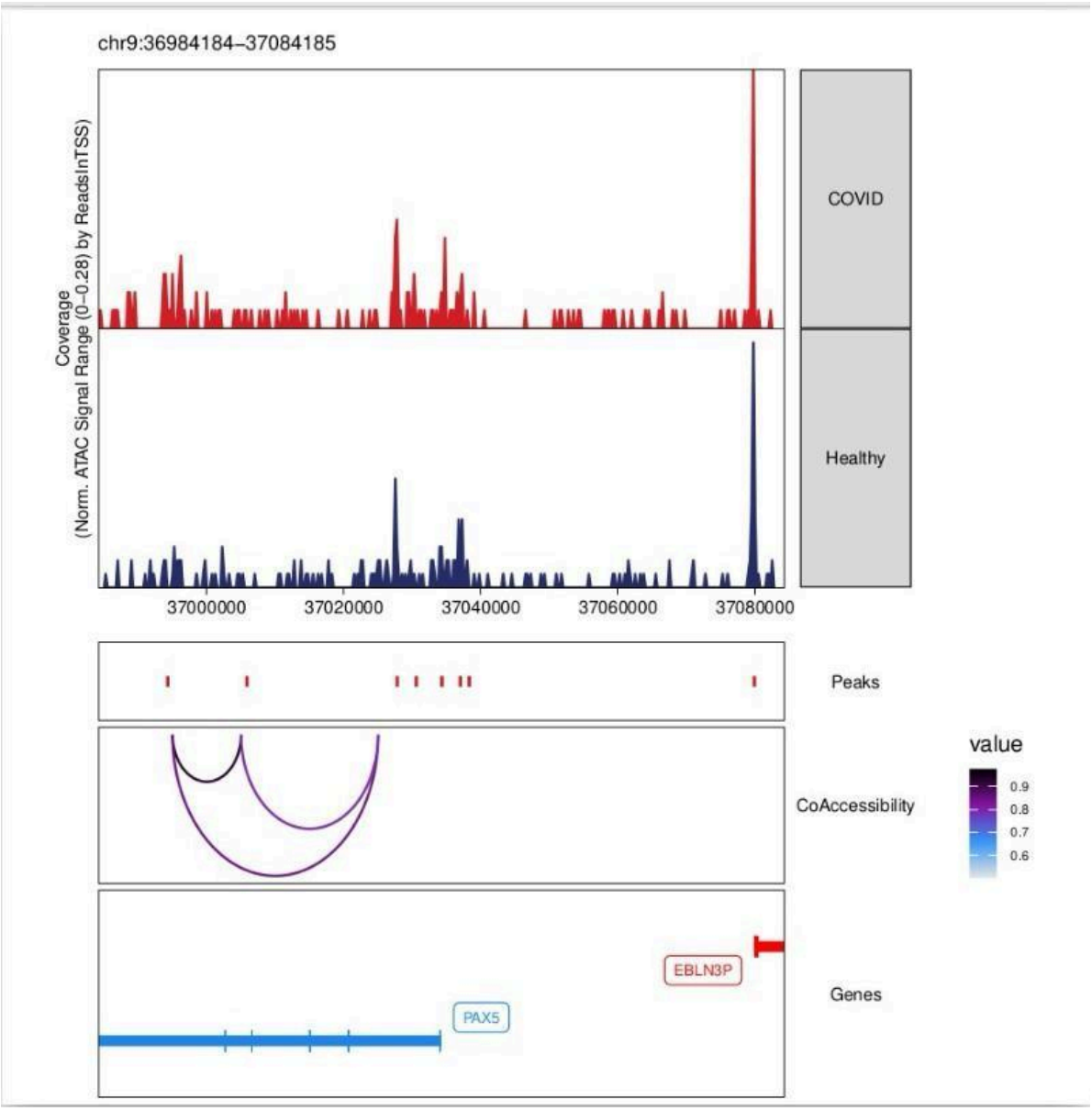
For CD34, the genome browser track shows a limited number of nearby accessible peaks, but no strong distal coaccessible regions linked to the transcription start site. This suggests that CD34 is primarily regulated by promoter proximal elements, with no clear distal enhancers detected within the analyzed window.

Number of potential enhancers for CD34: 0



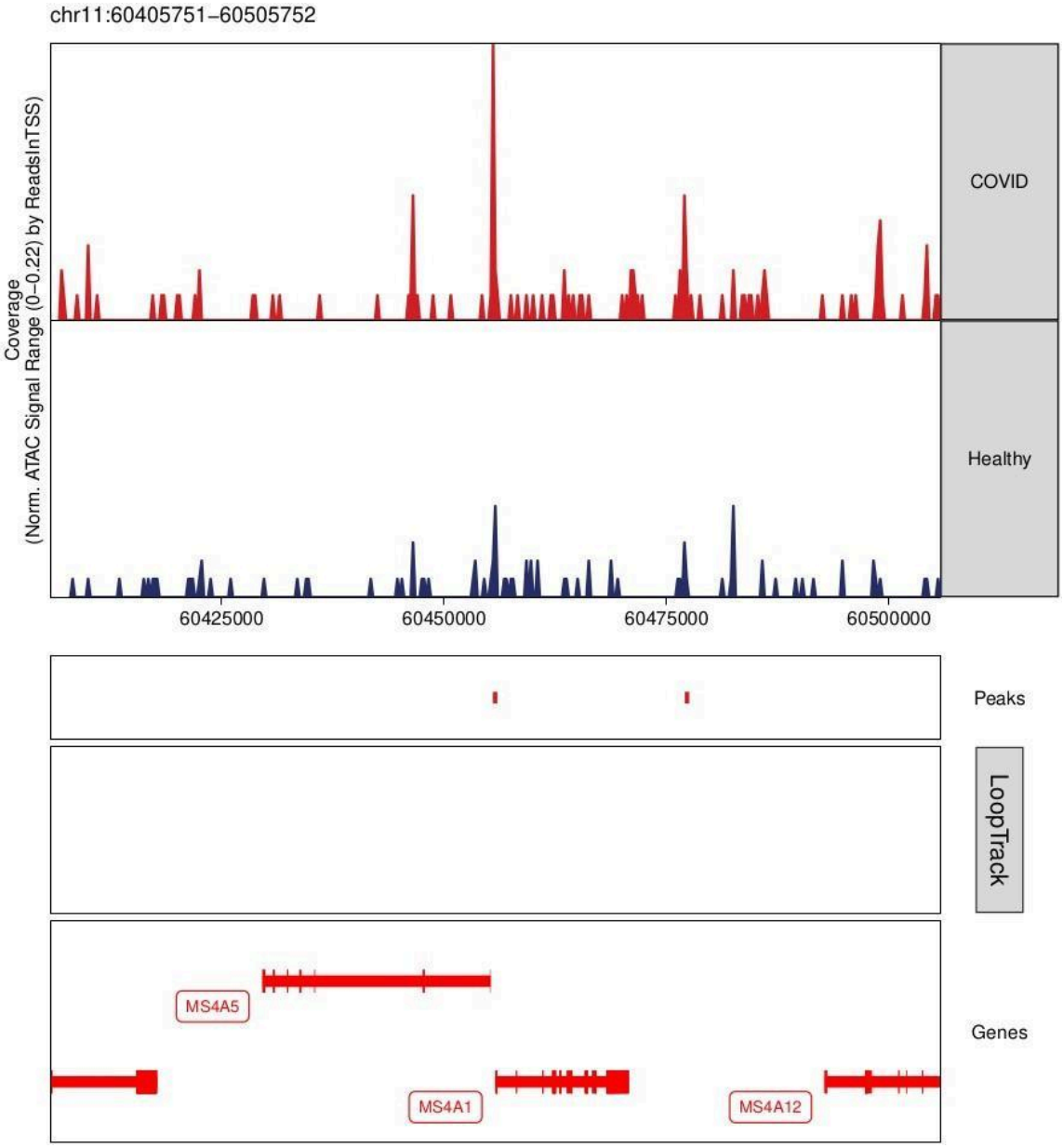
The genome browser track for GATA1 shows a strong promoter associated accessibility peak in both COVID and Healthy samples. There are only a few adjacent peaks and no distant coaccessible regions that connect to the GATA1 transcription start point. This suggests that promoter proximal elements, not distal enhancers, are the main regulators of GATA1.

Number of potential enhancers for GATA1: 0



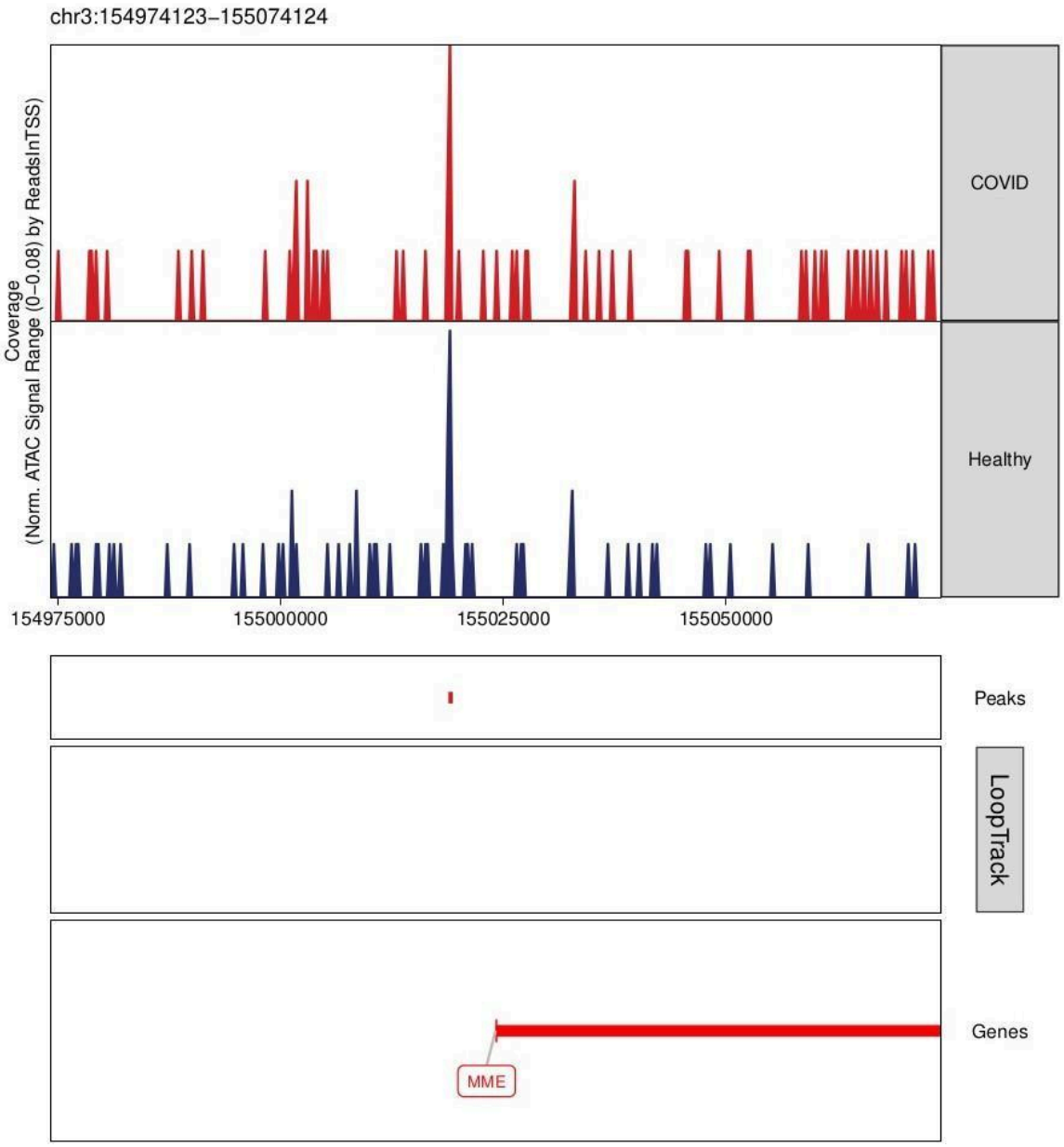
The PAX5 transcription start site is substantially coaccessible with several distal accessible peaks found in the PAX5 locus. Long range regulatory contacts are indicated by the presence of multiple high confidence coaccessibility arcs, which are compatible with enhancer mediated control of PAX5 expression.

Number of potential enhancers for PAX5: 2



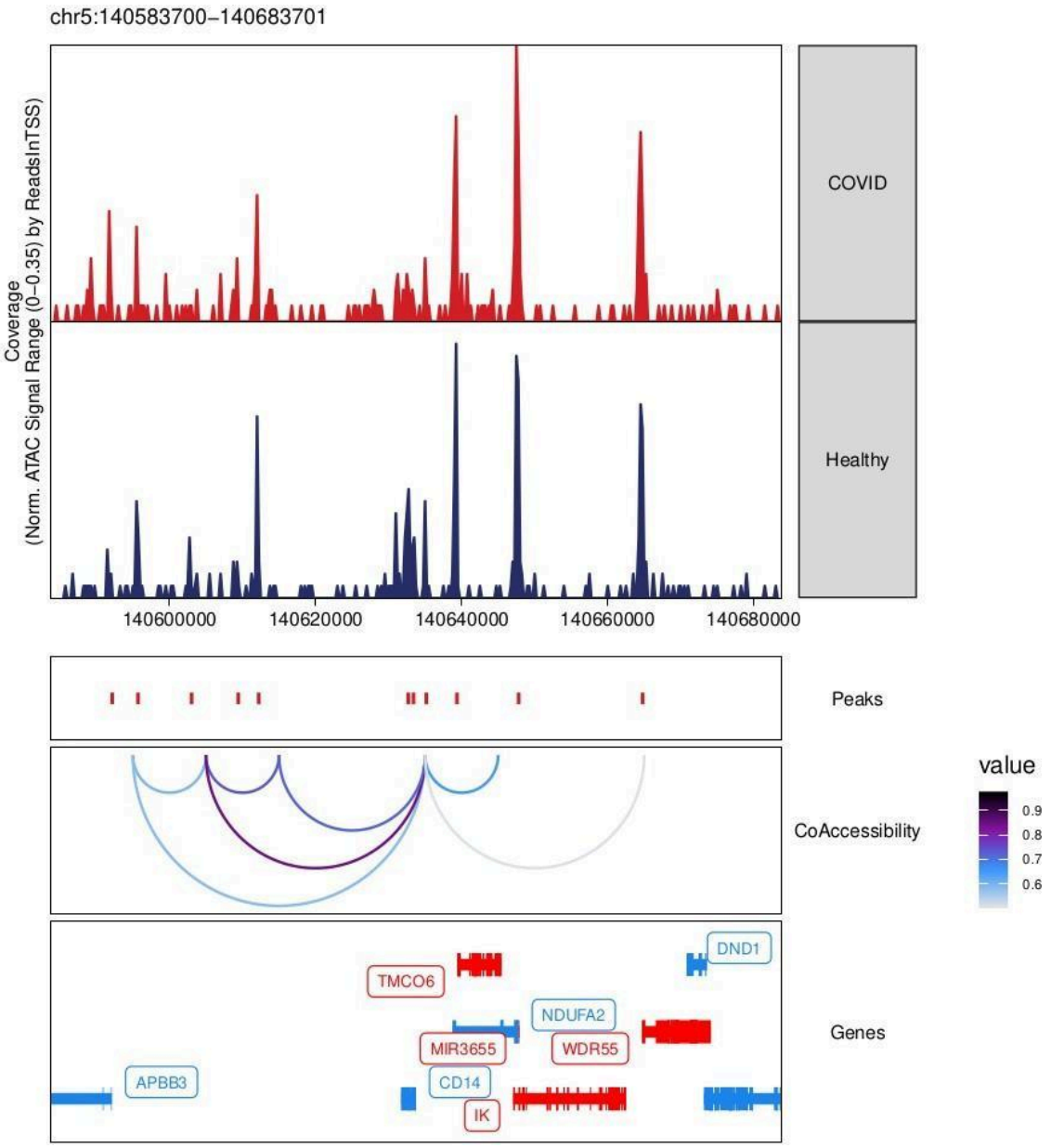
In both COVID and healthy samples, the MS4A1 gene has a robust promoter associated accessibility peak. There are very few neighbouring peaks and no distal coaccessible areas that connect to the MS4A1 transcription start point. This implies that promoter proximal elements, not distal enhancers, are primarily responsible for regulating MS4A1.

Number of potential enhancers for MS4A1: 0



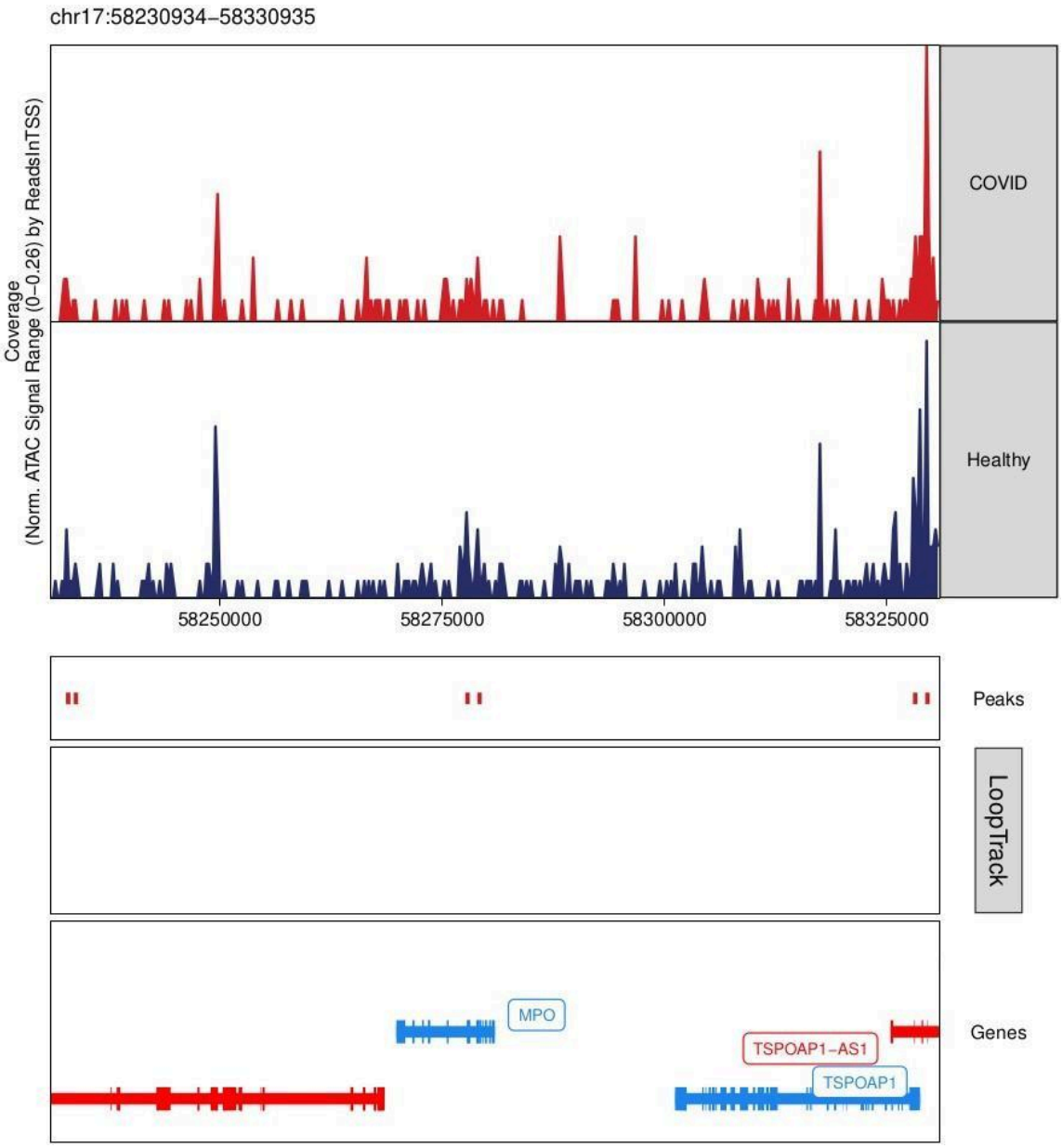
In both COVID and healthy samples, the MME locus has a single significant promoter associated accessibility peak. MME regulation is dominated by promoter proximal elements without discernible distal enhancers, as no distal accessible regions exhibit coaccessibility with the MME transcription start site.

Number of potential enhancers for MME: 0



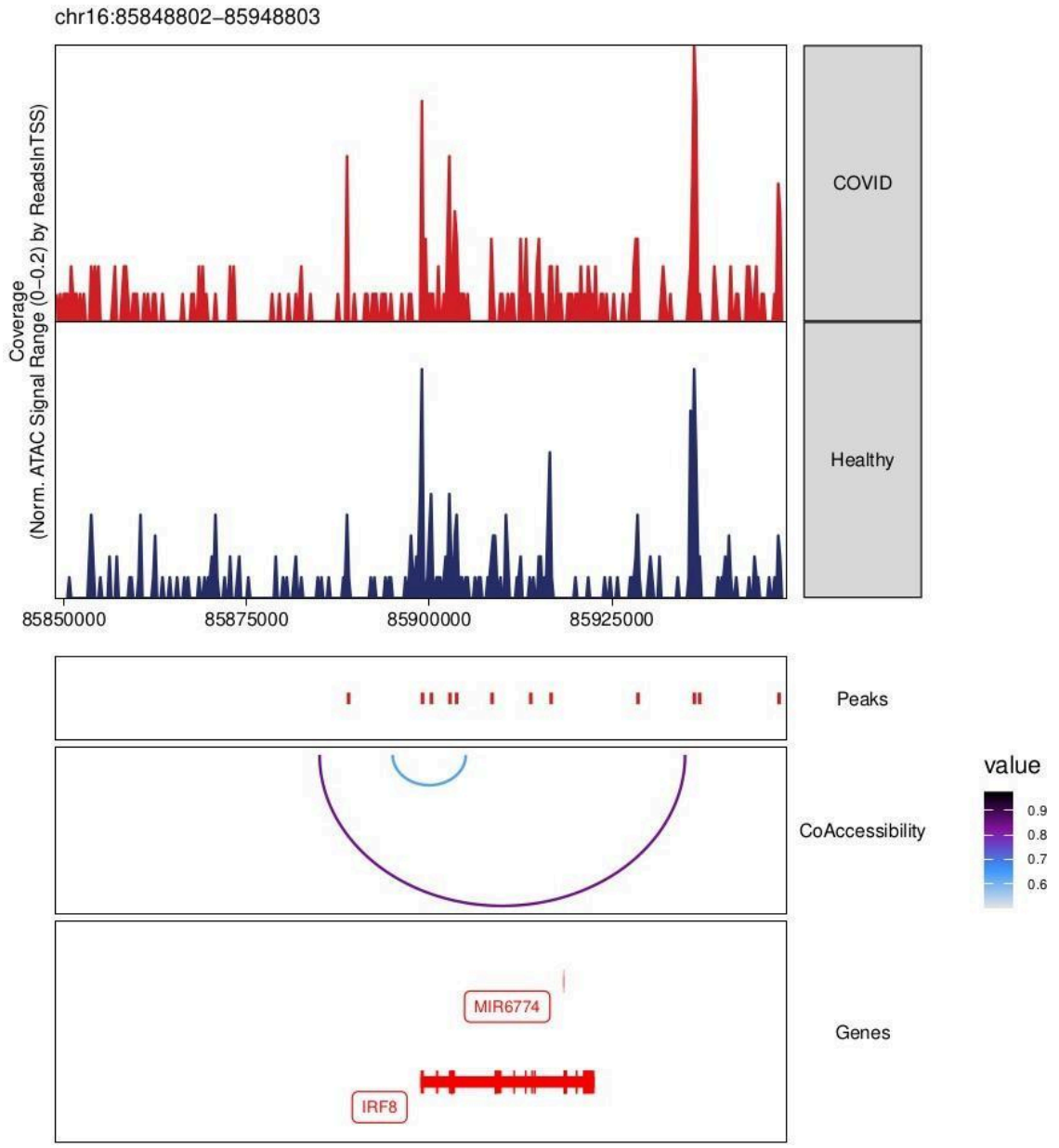
The CD14 locus has many distal accessible peaks as well as significant promoter accessibility. These distal peaks are connected to the CD14 transcription start site by many coaccessibility linkages, suggesting the existence of long range regulatory interactions. In the COVID situation, where accessibility is enhanced, these distal regions most likely constitute enhancer elements that contribute to CD14 regulation.

Number of potential enhancers for CD14: 2



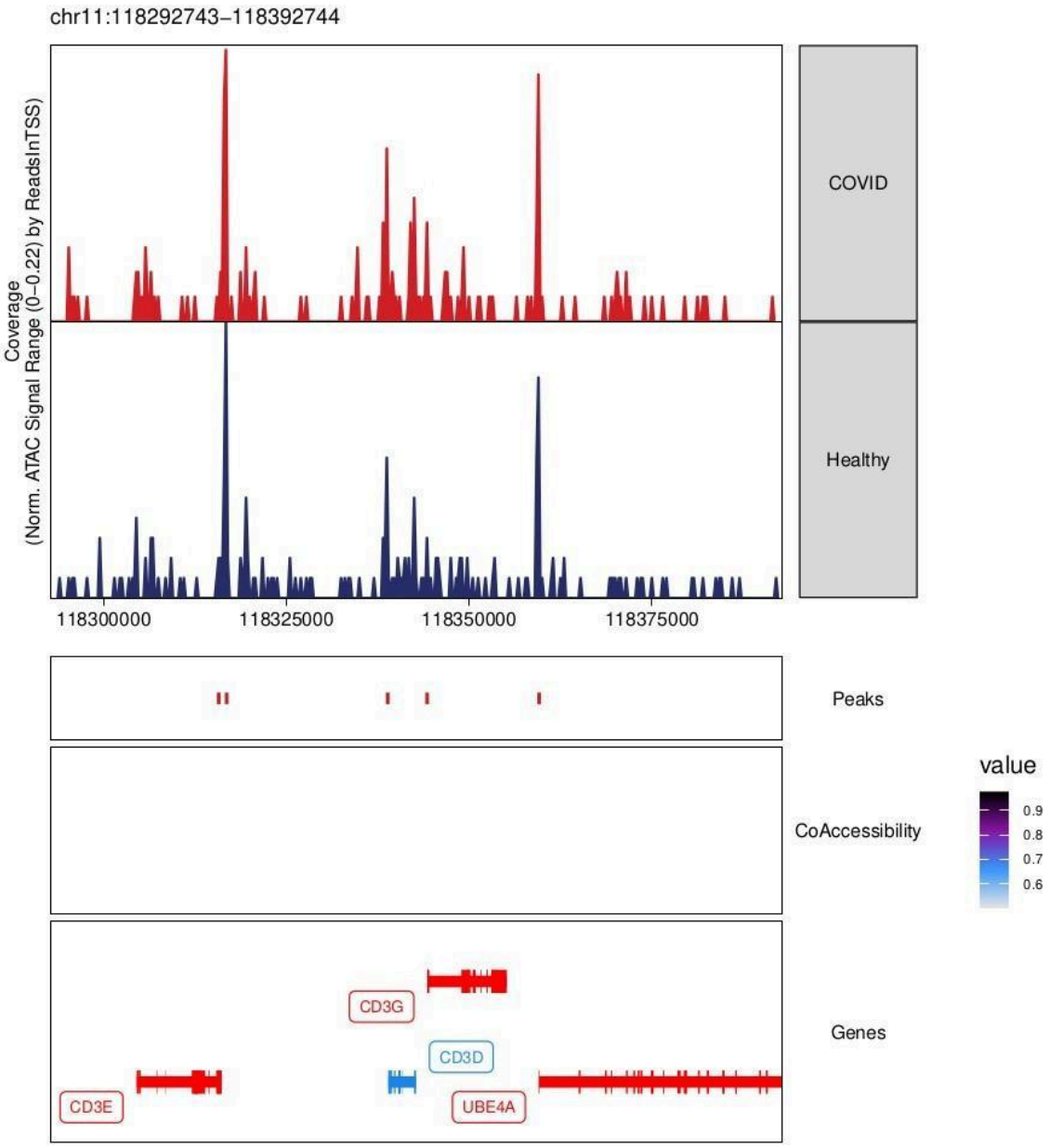
Strong promoter proximal accessibility is demonstrated by the MPO locus, especially in the COVID condition. Nevertheless, there are no discernible coaccessibility connections between the MPO transcription start site and distant peaks. This implies that there is no conclusive evidence for distal enhancer elements and that MPO control in this sample is mostly promoter driven.

Number of potential enhancers for MPO: 0



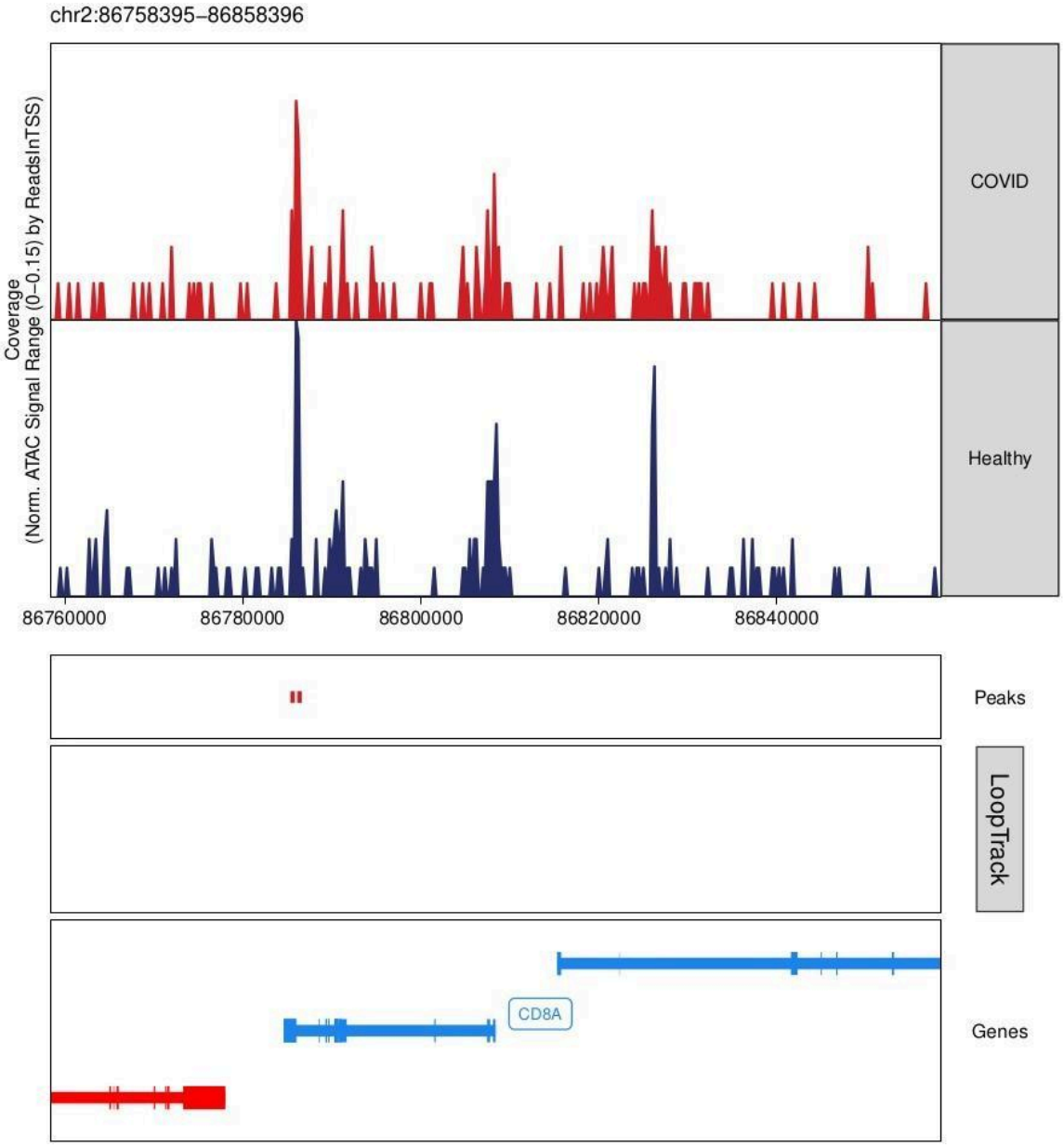
The IRF8 transcription start site is considerably coaccessible with several distal ATACseq peaks found in the IRF8 gene. There are two different coaccessibility linkages found, suggesting the existence of distal regulatory elements that probably serve as enhancers. This implies that several long range enhancer promoter interactions control IRF8 expression.

Number of potential enhancers for IRF8: 2



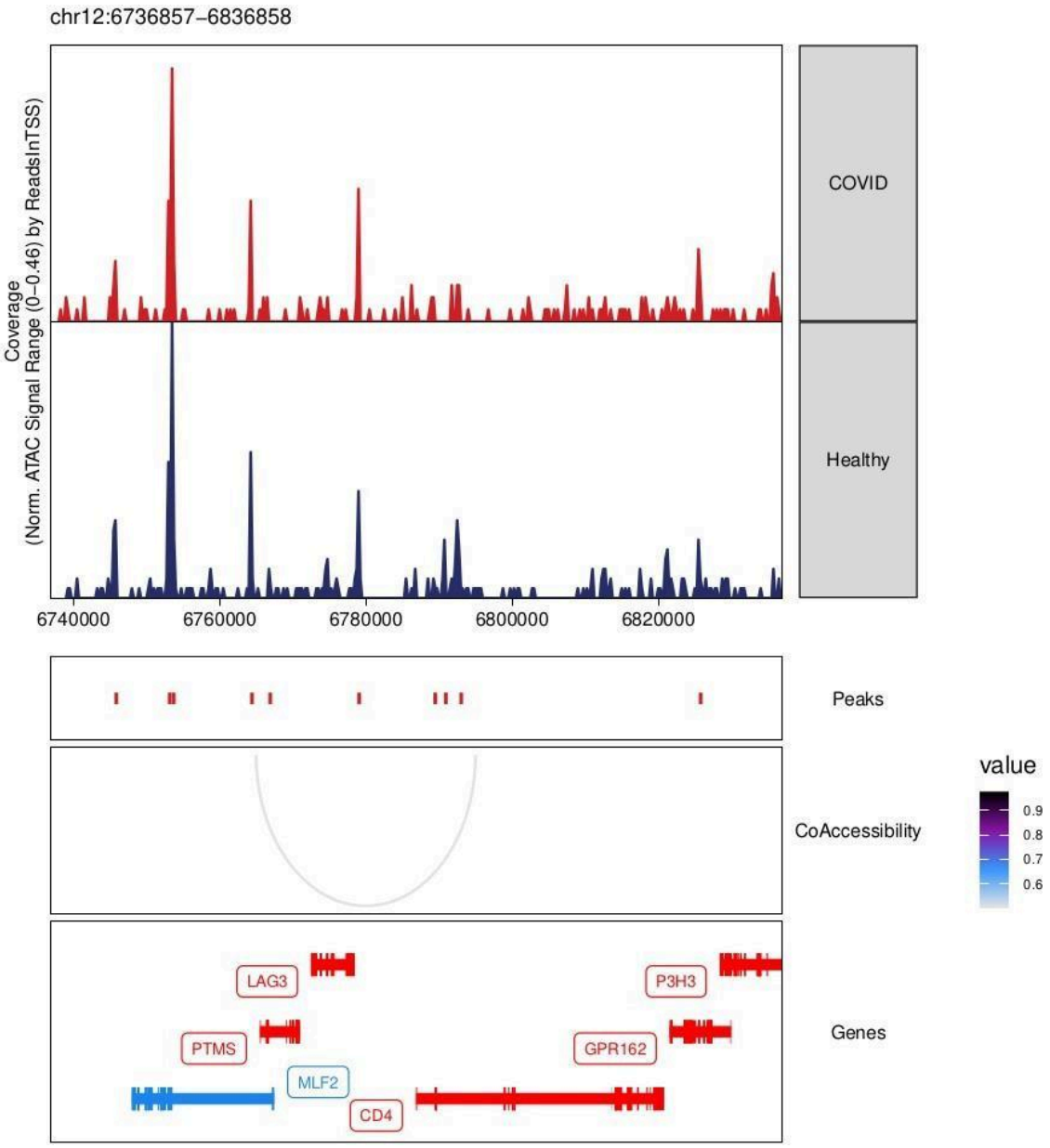
Both COVID and healthy samples have several accessible chromatin peaks at the CD3 gene. Nevertheless there are no discernible coaccessibility connections between the CD3D/CD3G transcription start sites and distal peaks. Since no distal enhancers were found under the applied coaccessibility threshold, it appears that the control of CD3 genes in this dataset is mostly promoter proximal.

Number of potential enhancers for CD3: 0



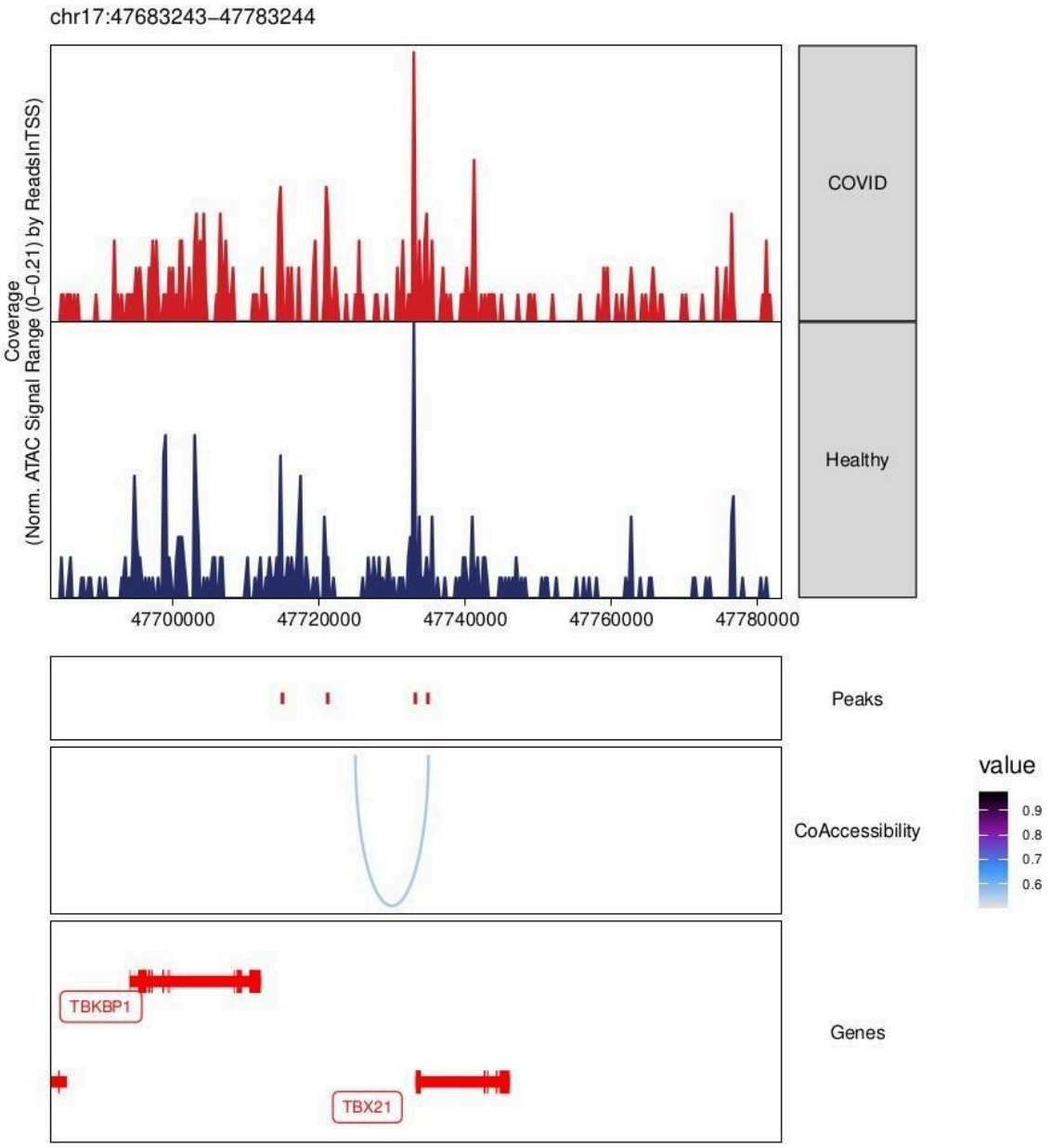
In both COVID and healthy samples, the CD8A locus has significant promoter associated chromatin accessibility. Nevertheless there are no discernible coaccessibility connections between the CD8A transcription start site and distal peaks. With no distal enhancers found, this implies that CD8A regulation in this sample is mostly promoter driven.

Number of potential enhancers for CD8A: 0



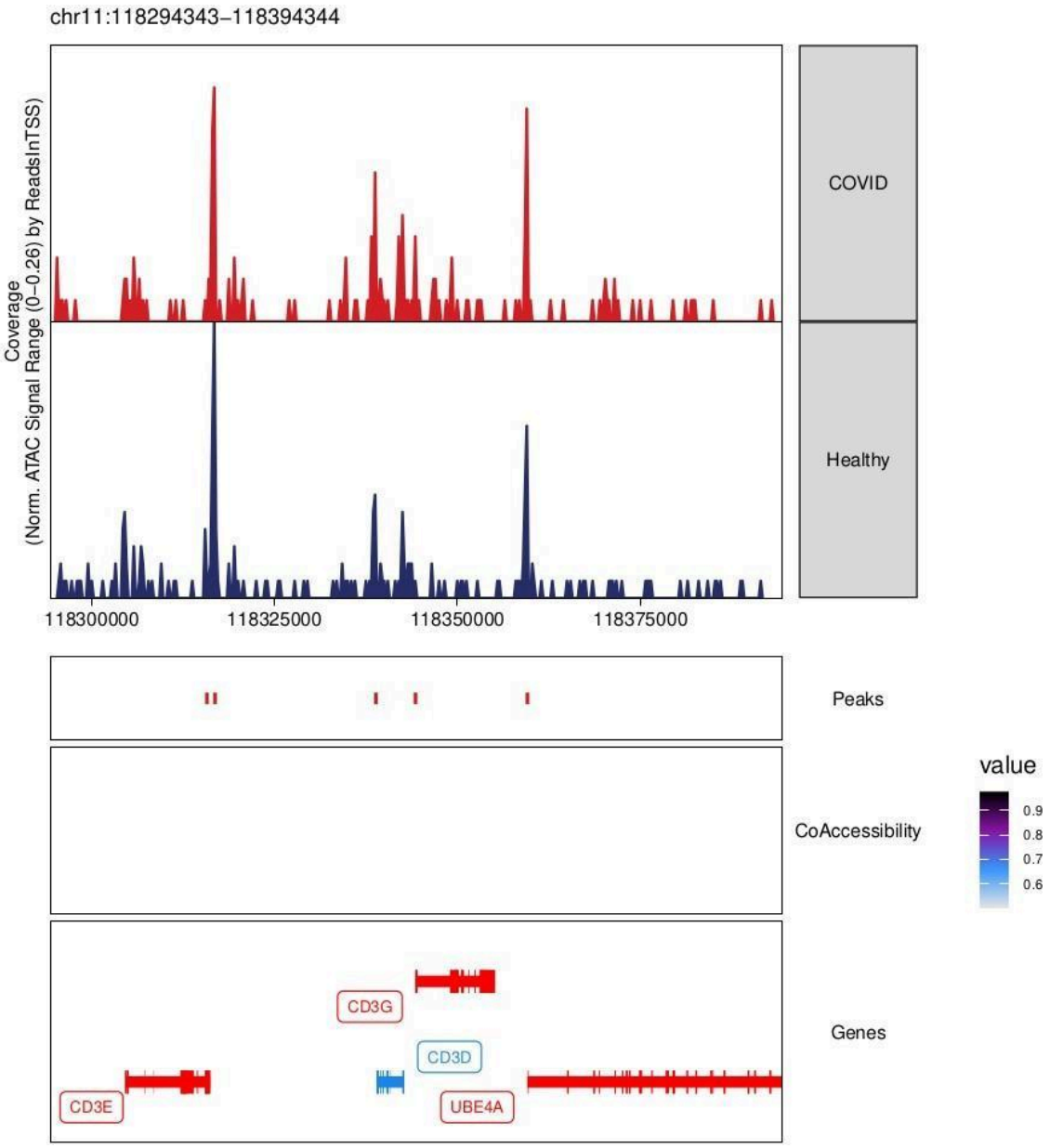
In both COVID and healthy samples, the CD4 region exhibits high chromatin accessibility. There are several distal accessible peaks, one of which has a strong coaccessibility connection to the CD4 transcription start point. This implies that CD4 regulation is influenced by a distant regulatory factor.

Number of potential enhancers for CD4: 1



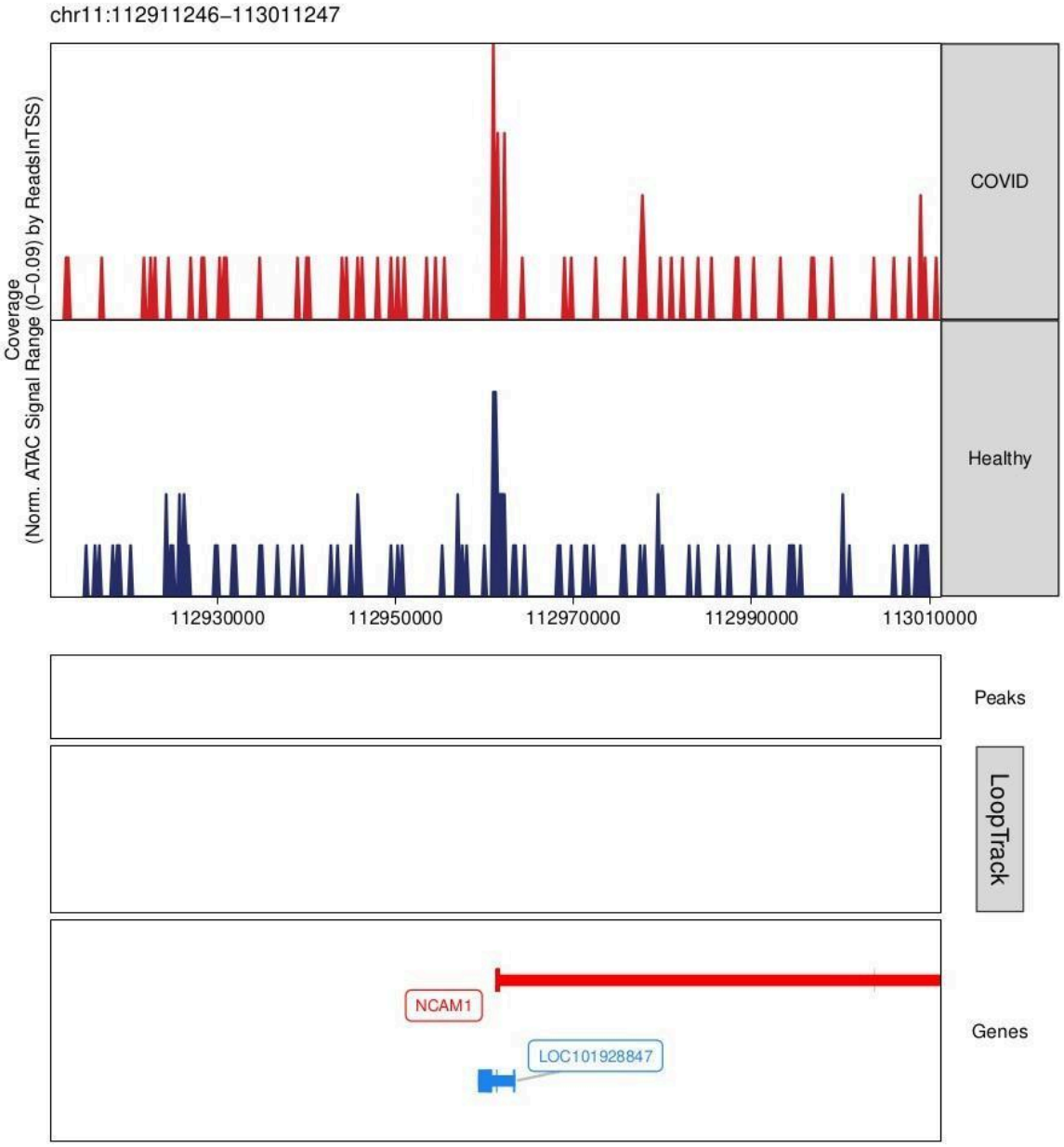
Both COVID and healthy samples exhibit distinct chromatin accessibility at the TBX21 gene. Significant coaccessibility between a distal accessible peak and the TBX21 transcription start site suggests a regulatory link. This implies that TBX21 expression may be regulated by a distal enhancer.

Number of potential enhancers for TBX21: 1



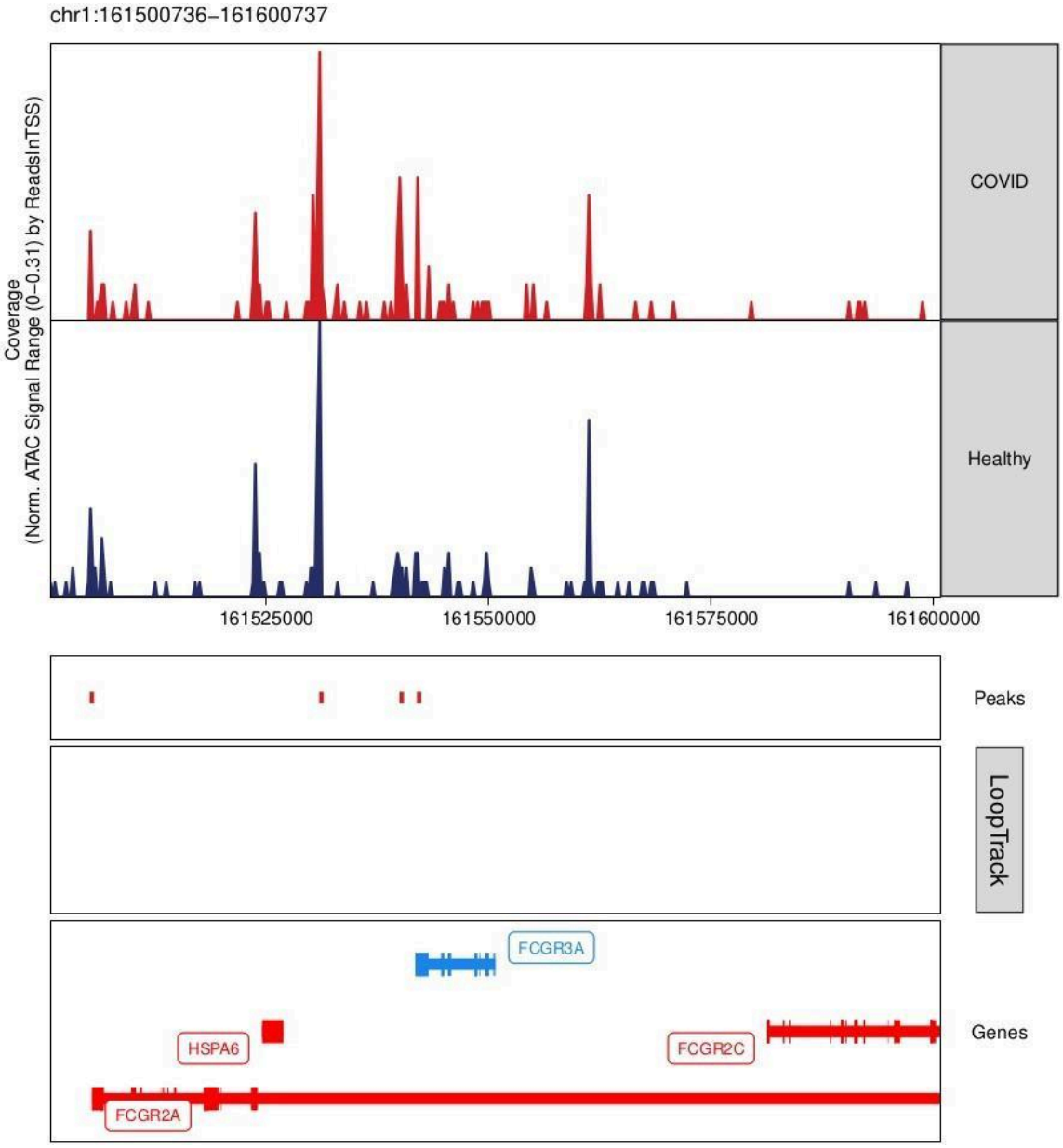
In both COVID and healthy samples, the CD3G gene has several accessible chromatin peaks. Nevertheless there are no discernible coaccessibility connections between the CD3G transcription start site and distal peaks. As a result, this research does not allow for the confident assignment of any distal regulatory enhancers to CD3G.

Number of potential enhancers for CD3G: 0



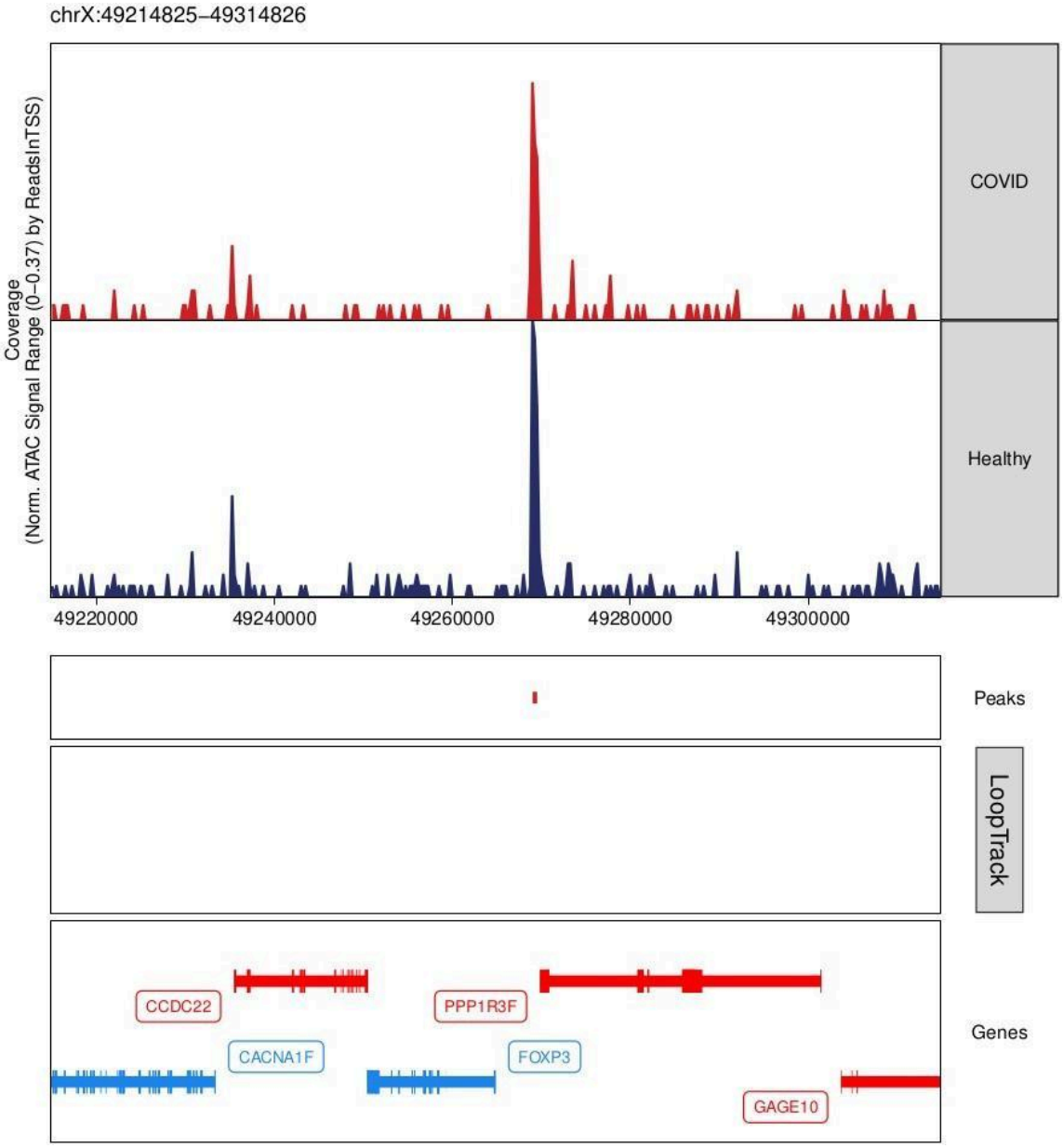
In both COVID and healthy samples, the NCAM1 locus has accessible chromatin close to the gene body. Nevertheless there are no discernible distal peaks or strong coaccessibility connections to the NCAM1 transcription start point. Therefore this research is unable to identify any possible enhancers for NCAM1.

Number of potential enhancers for NCAM1: 0



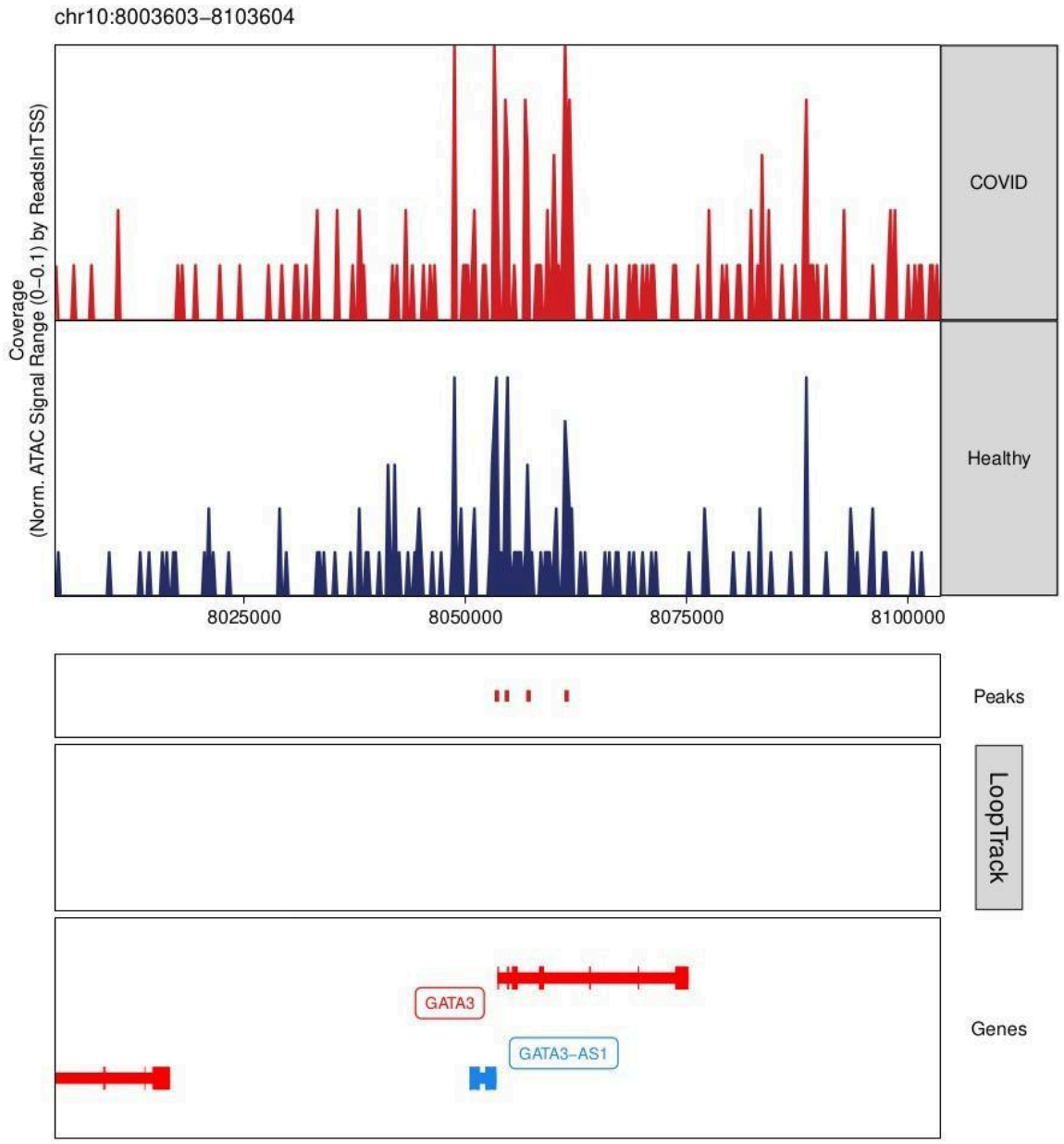
In both COVID and healthy samples, the FCGR3A locus exhibits accessible chromatin and multiple adjacent peaks. Nevertheless there are no discernible coaccessibility connections between the FCGR3A TSS and distal peaks. As a result this analysis does not reveal any possible enhancers for FCGR3A.

Number of potential enhancers for FCGR3A: 0



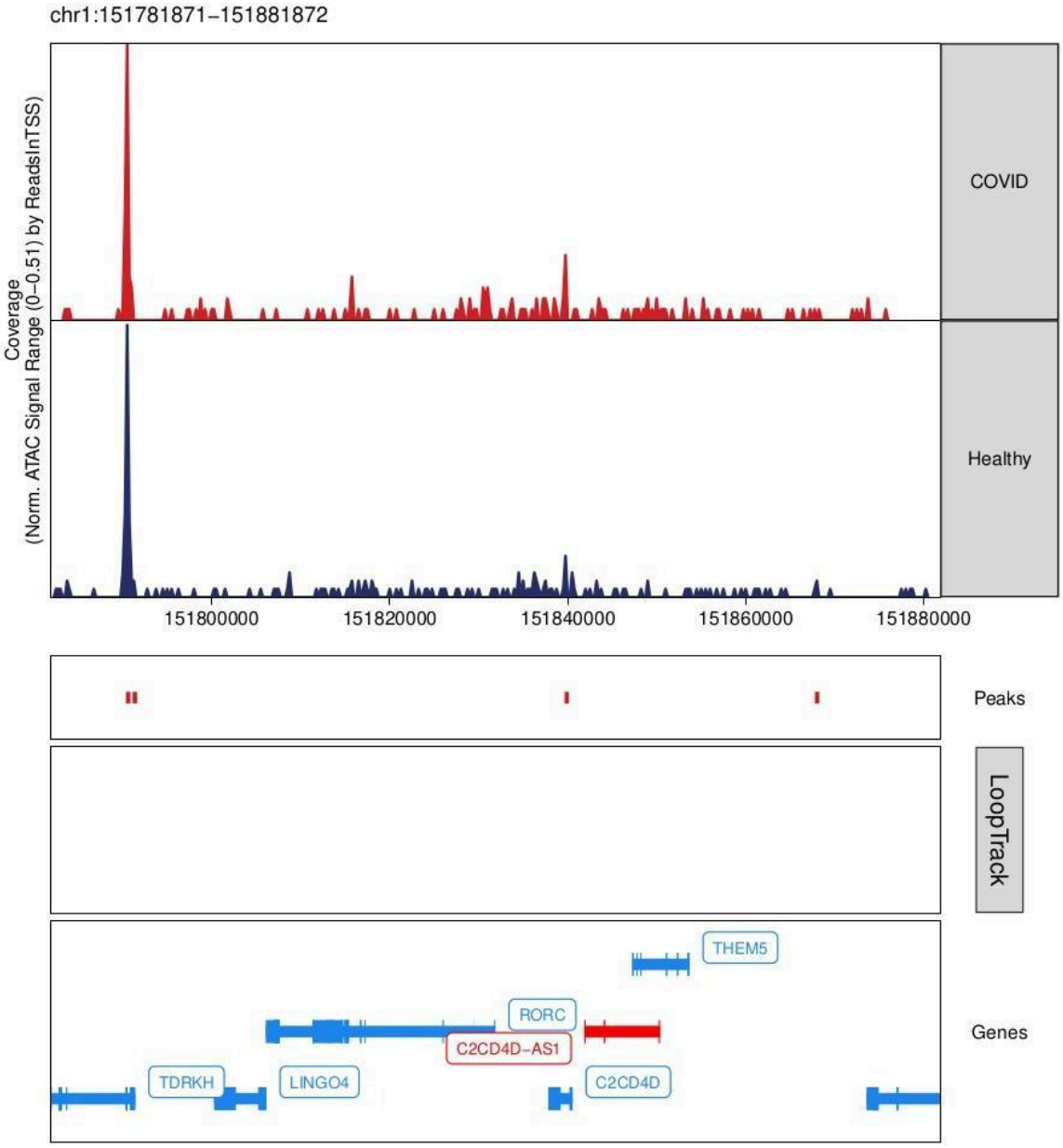
Both COVID and healthy samples exhibit a robust accessible promoter peak at the FOXP3 locus. Nevertheless there are no discernible coaccessibility connections between the FOXP3 transcription start site and distal peaks. Consequently this analysis does not reveal any possible enhancers for FOXP3.

Number of potential enhancers for FOXP3: 0



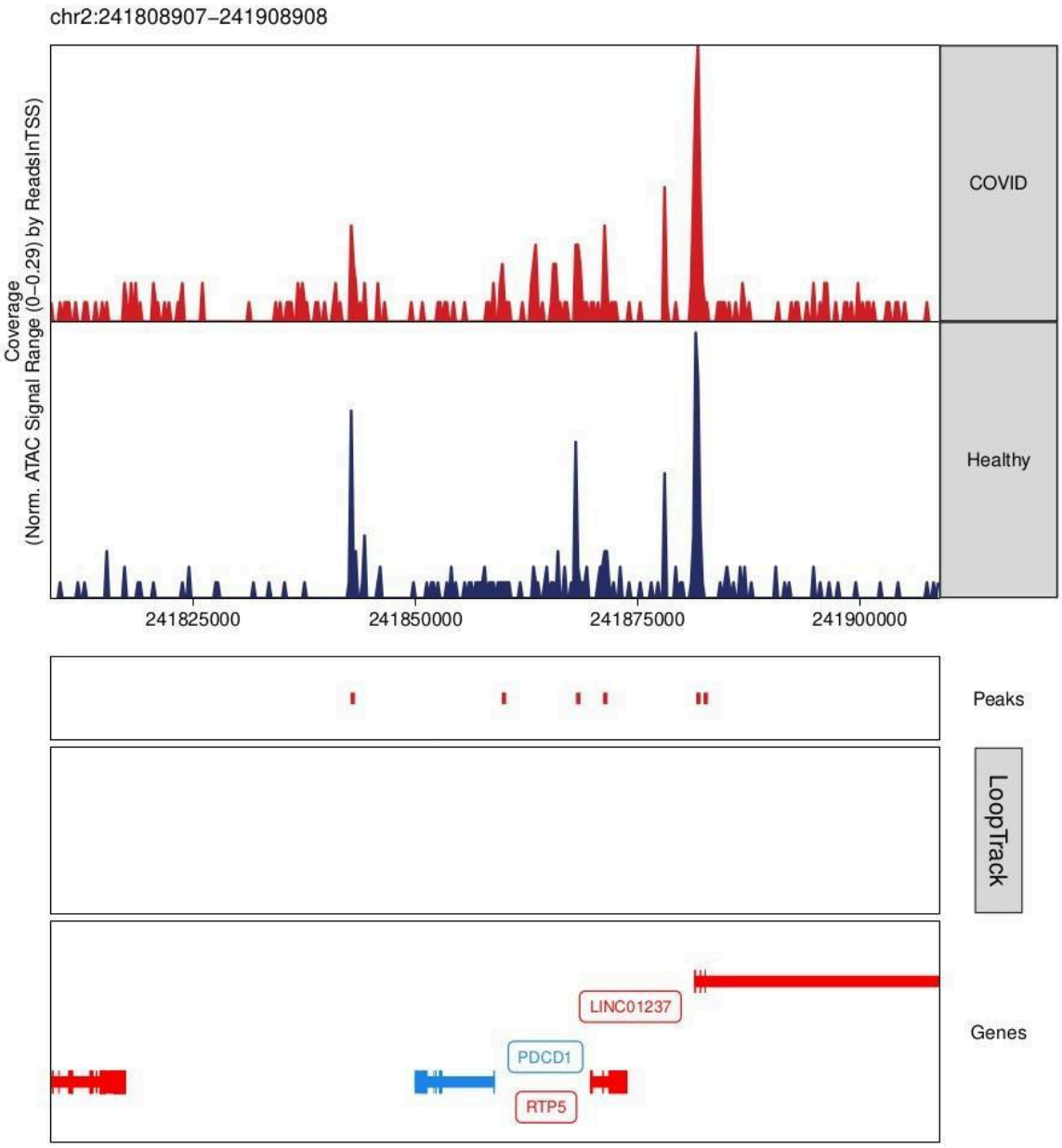
In both COVID and healthy samples, the GATA3 locus exhibits several accessible chromatin regions. Nevertheless there are no discernible coaccessibility connections between the GATA3 transcription start site and distal peaks. Consequently this analysis does not identify any potential enhancers for GATA3.

Number of potential enhancers for GATA3: 0



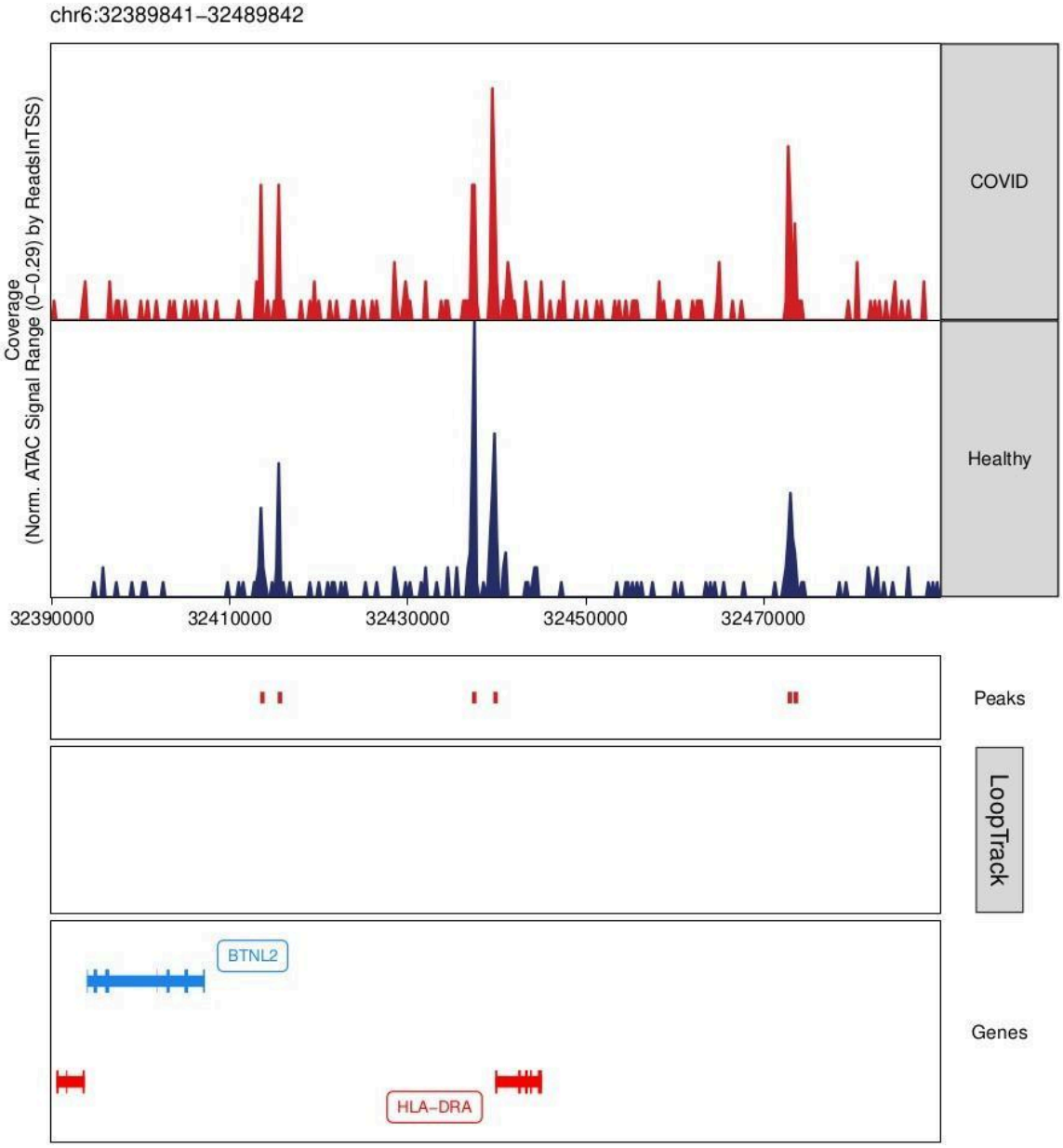
Both COVID and healthy samples exhibit strong promoter accessibility at the RORC locus, with multiple accessible peaks in the vicinity. Nevertheless there are no discernible coaccessibility connections between the RORC transcription start site and distal peaks. As a result this analysis does not identify any possible enhancers for RORC.

Number of potential enhancers for RORC: 0



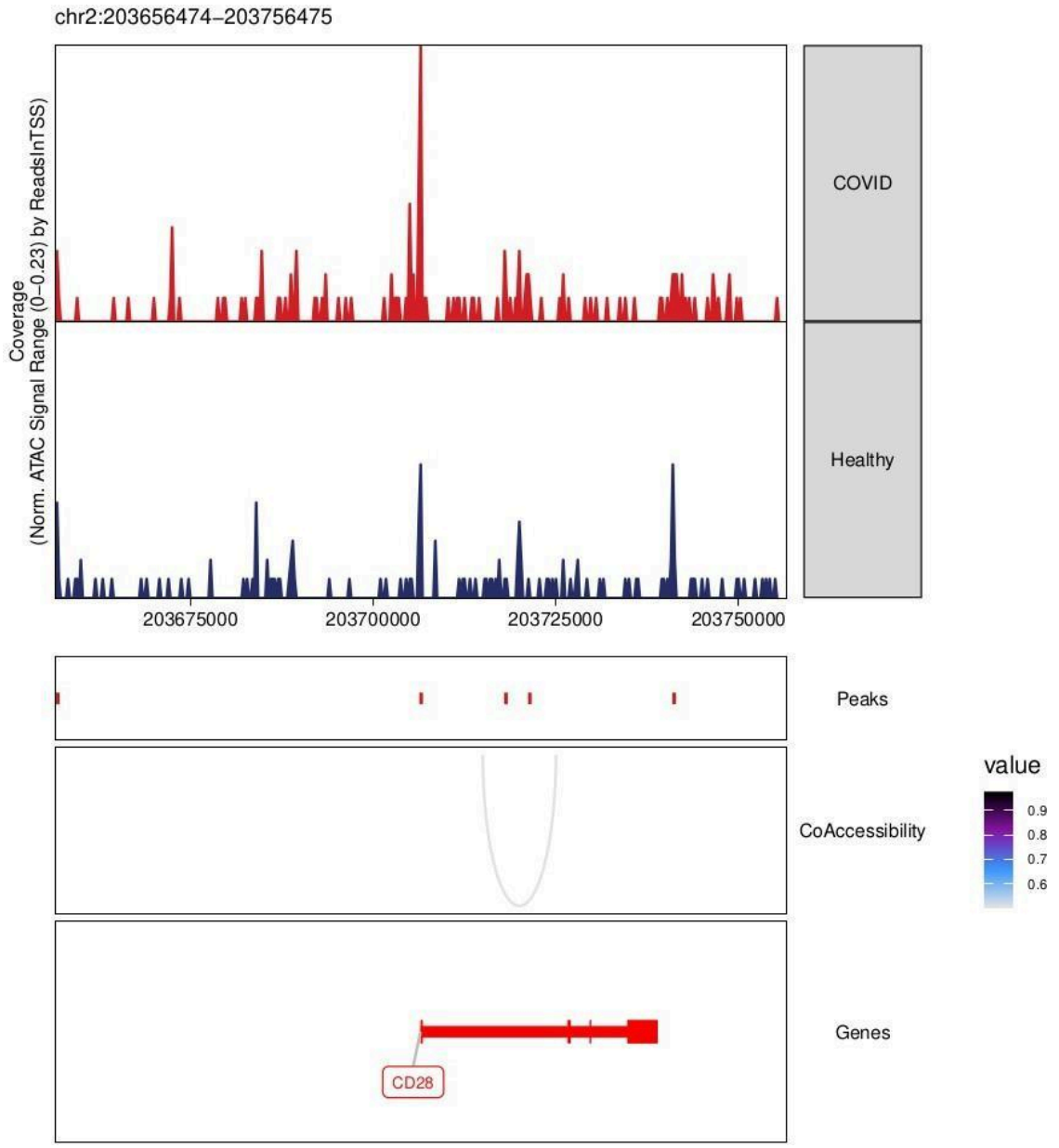
In both COVID and healthy samples, the PDCD1 locus exhibits multiple distal accessible peaks and strong promoter accessibility. Nevertheless there are no discernible coaccessibility connections between these peaks and the PDCD1 transcription start site. Consequently no potential enhancers can be identified for PDCD1 using the applied coaccessibility criteria.

Number of potential enhancers for PDCD1: 0



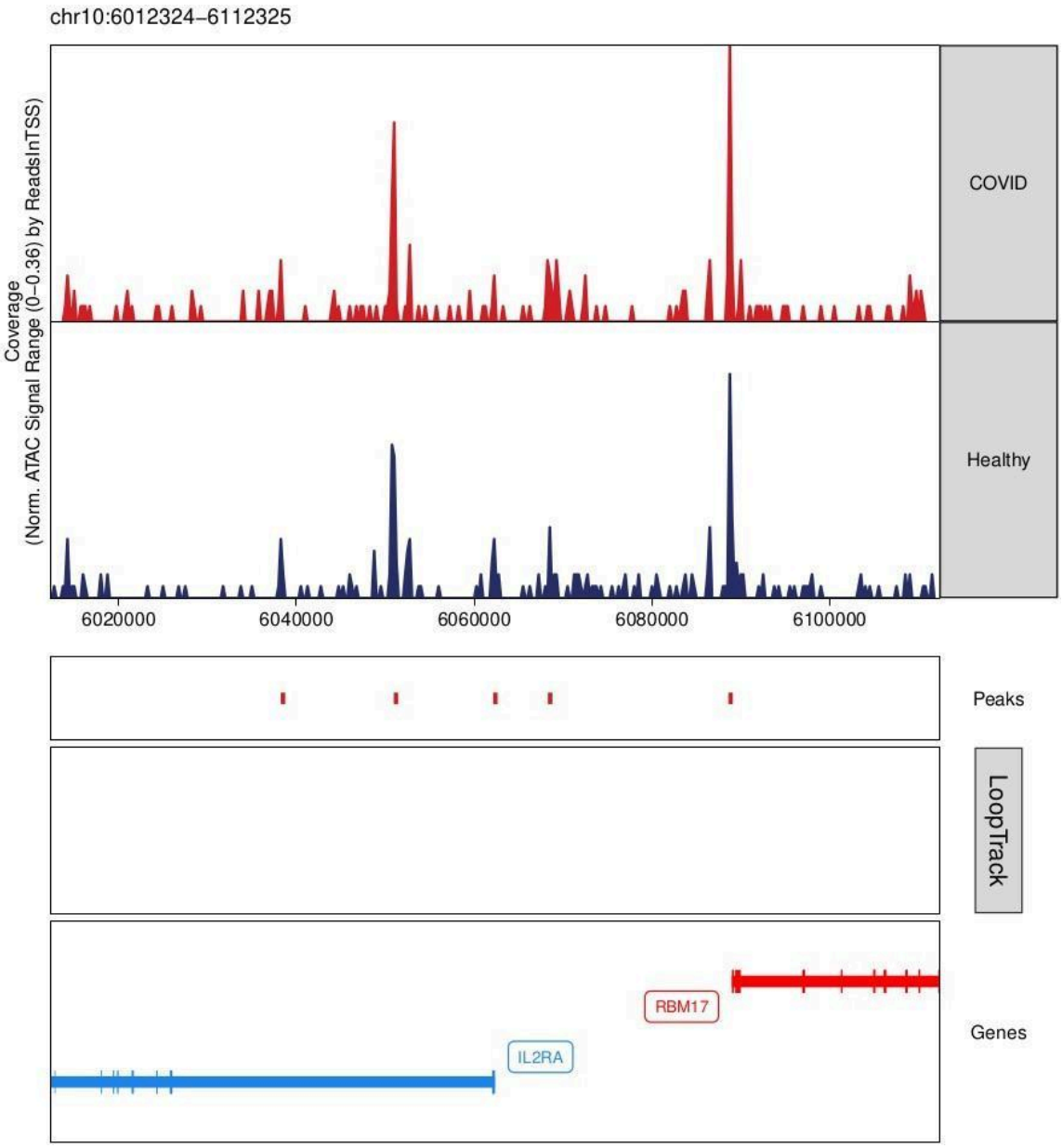
In both COVID and healthy samples, the HLA-DRA locus exhibits strong promoter accessibility, with multiple distal accessible peaks. Nevertheless there are no discernible coaccessibility connections between the HLA-DRA transcription start site and these distal regions. Therefore coaccessibility analysis does not reveal any potential enhancers for HLA-DRA.

Number of potential enhancers for HLA-DRA: 0



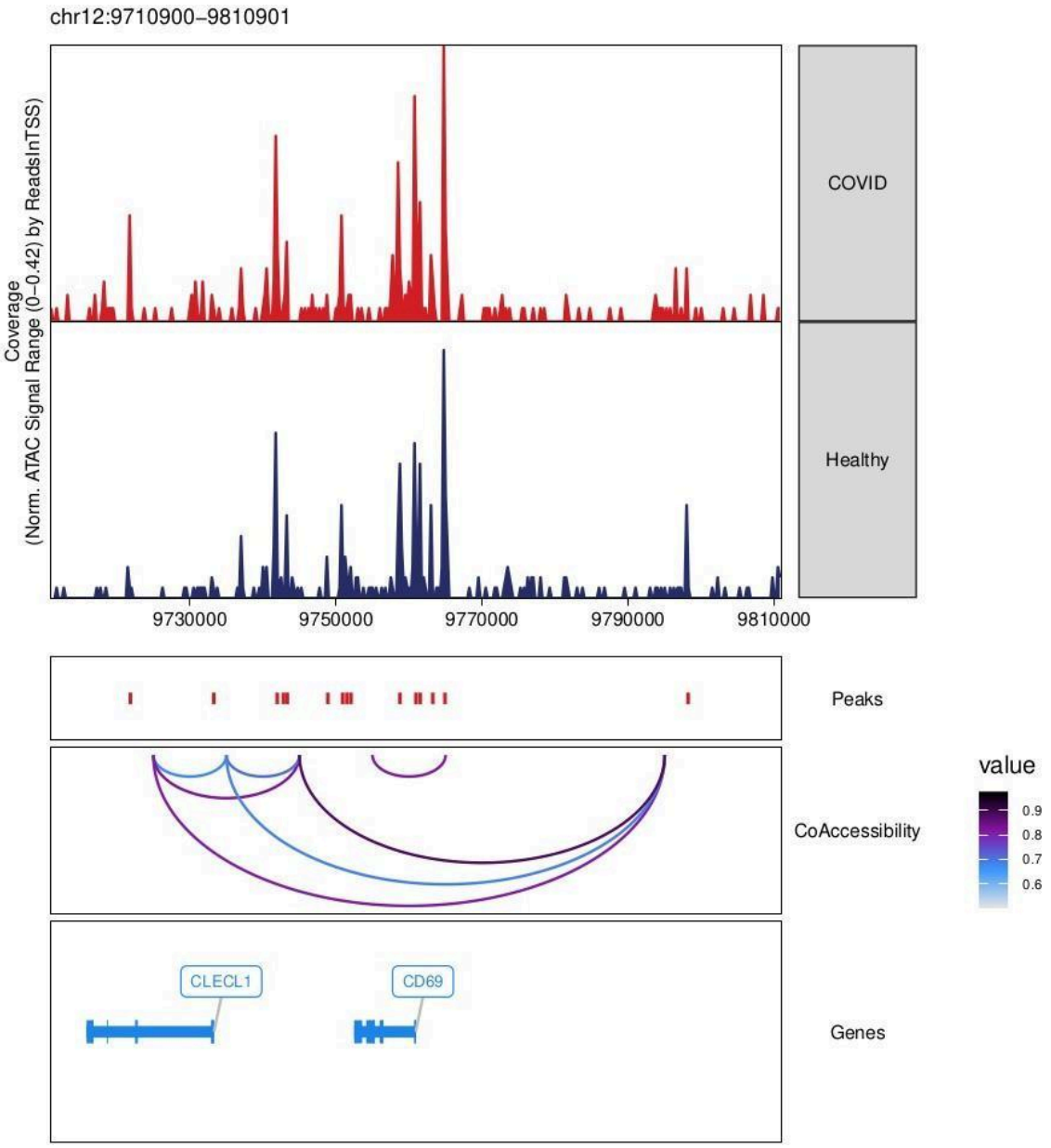
The CD28 locus shows strong promoter accessibility in both COVID and healthy samples. Several distal accessible peaks are present within +/-50 kb of the gene. A distal peak shows considerable coaccessibility with the transcription start site of CD28, suggesting a possible enhancer that modulates CD28 expression.

Number of potential enhancers for CD28: 1



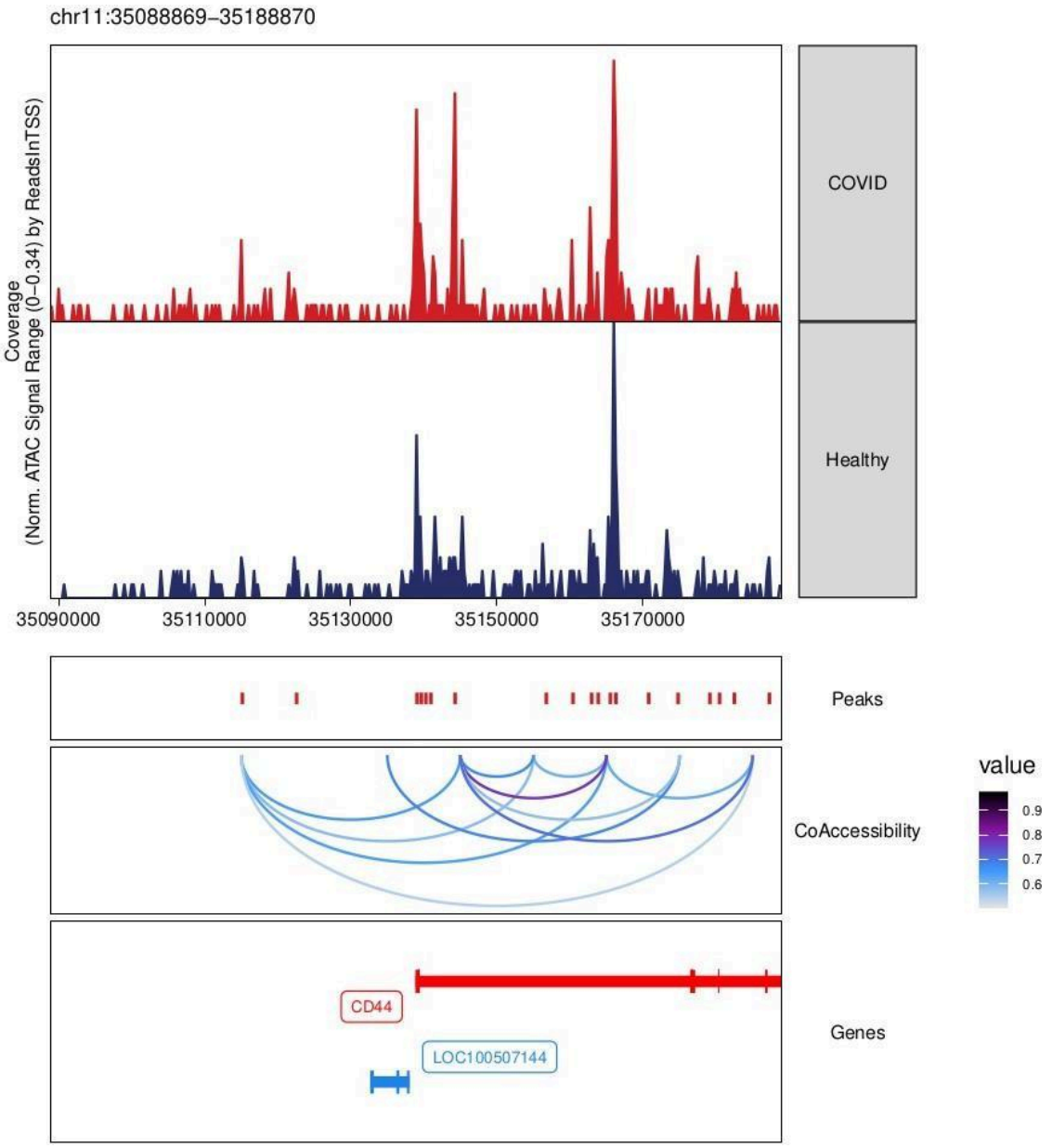
The IL2RA region exhibits a distinctly open promoter in samples from both COVID patients and healthy individuals, along with several additional nearby accessible distal regions. Nonetheless there are no noteworthy coaccessibility connections detected between the distal peaks and the transcription start site of IL2RA. Consequently it is not possible to identify any potential enhancers for IL2RA based on coaccessibility.

Number of potential enhancers for IL2RA: 0



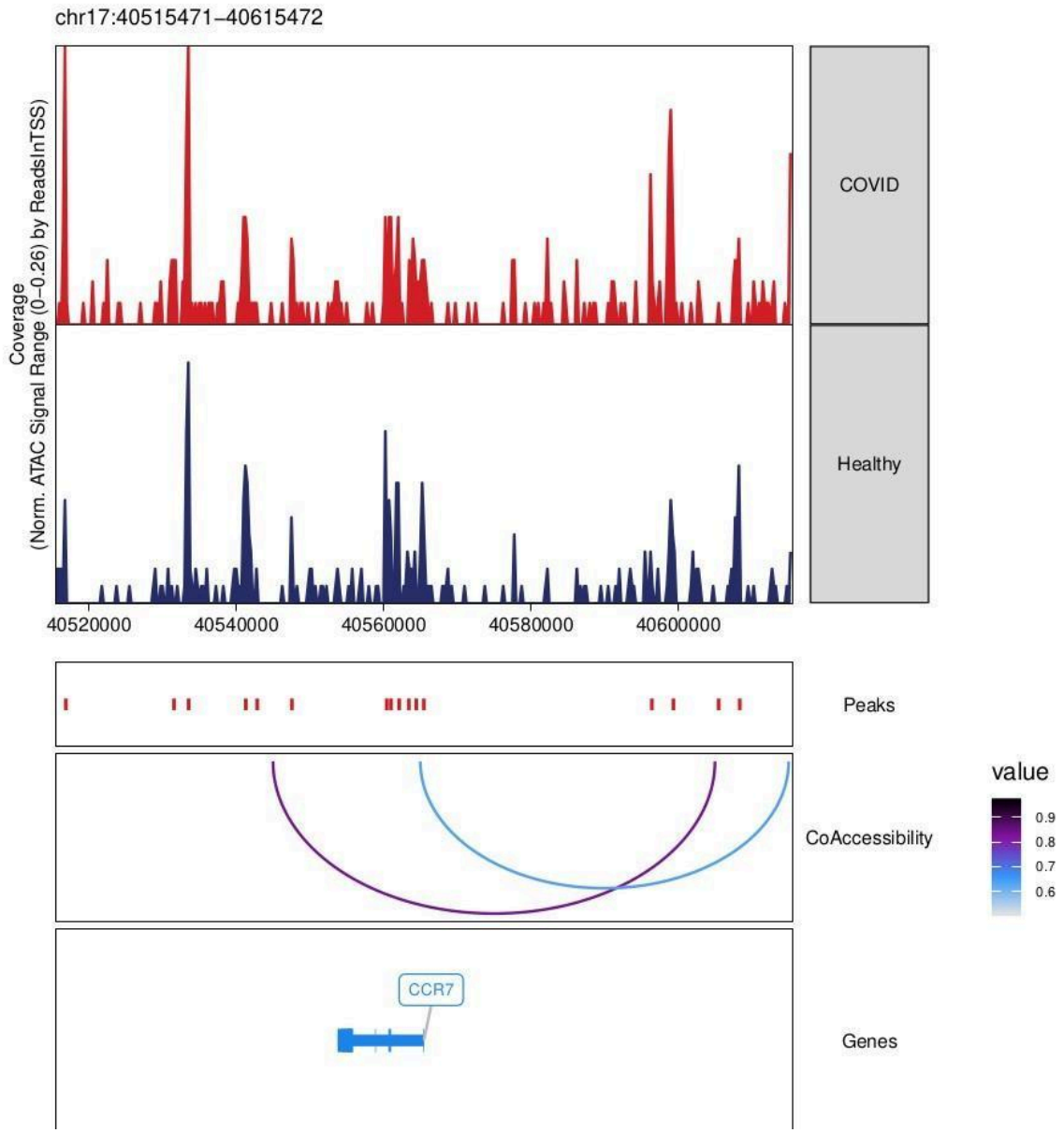
The CD69 region exhibits notable promoter accessibility along with various distant accessible areas. Several distal peaks demonstrate significant coaccessibility with the transcription start site of CD69, suggesting that multiple potential enhancers are likely involved in the regulation of CD69.

Number of potential enhancers for CD69: 3



There are many distal accessible areas and significant chromatin accessibility at the promoter of the CD44 gene. Significant coaccessibility between numerous distal peaks and the CD44 transcription start point suggests that multiple putative enhancers are probably controlling CD44 expression.

Number of potential enhancers for CD44: 5



There are multiple distal accessible areas and accessible chromatin at the promoter of the CCR7 gene. There may be putative enhancer elements controlling CCR7 expression because two distal peaks exhibit strong coaccessibility with the CCR7 promoter.

Number of potential enhancers for CCR7: 2

References:

https://www.archrproject.com/bookdown/index.html

https://ucdavis-bioinformatics-training.github.io/2022-July-Advanced-Topics-in-Single-Cell-RNA-Seq-ATAC/data_analysis/scATAC_analysis

https://github.com/GreenleafLab/ArchR